

How Much Does Fusion Design Matter? Partition, Seed, and Cluster-Aware Evaluation of Hybrid LSTM–GAT Traffic Anomaly Detectors

Ayhan Gültekin¹ | Kerem Kartal² | Ahmet Yusuf Şimşek³ | Rümeyza Yeşilova³ | Ahmet Sayar⁴

¹Student Affairs Department, Kocaeli University, Kocaeli, Türkiye | ²Department of Software Engineering, Kocaeli University, Kocaeli, Türkiye | ³Department of Artificial Intelligence and Machine Learning, Kocaeli University, Kocaeli, Türkiye | ⁴Department of Artificial Intelligence Engineering, Bahçeşehir University, Istanbul, Türkiye

Correspondence: Ayhan Gültekin (ayhan.gultekin@kocaeli.edu.tr)

Abstract—Hybrid spatio-temporal architectures for traffic anomaly detection are typically reported through a single training run on a single data partition, with overlapping sliding windows treated as independent observations. This paper asks how much of such evidence survives a controlled protocol. Using a physics-informed synthetic trajectory dataset over an OpenStreetMap-derived road network, we hold the task, optimization budget, checkpoint criterion, and threshold rule fixed; vary one component at a time across nine encoder configurations, four fusion operators, and two graph constructions; and report every result over multiple seeds. Because windows are cut with stride 10 from length-50 sequences and therefore overlap by 80%, all inference uses the *trip* as the statistical unit, with cluster bootstrap, sign-flip permutation tests, and Holm correction. Five findings emerge. First, on the fixed partition—which contains 4,799 windows but only 14 trips—no comparison remains significant once the correct unit and multiplicity correction are applied; seven comparisons that are significant at the window level are not. Second, the fusion ranking depends on how many seeds are averaged: the sample-wise gate that motivated the original design ranks last of four on the fixed partition, first under cross-validation over two seeds, and fourth over five, while the temporal-only control—which has no spatial branch at all—ranks first. Three orderings arise from the same data, the same folds, and the same protocol. Third, the apparent architectural spread is degenerate runs rather than architecture. Over 125 cross-validated runs the four hybrids span 0.0582 F1, which exceeds a single run’s replication error of 0.0414; excluding the three degenerate runs the span is 0.0095 F1 and the within-configuration standard deviation is 0.037–0.044 for all five configurations alike. Fourth, that failure mode is a property of learning a mixing weight, not of the proposed operator. Across 500 repeated partitions it occurs in 8 runs, every one belonging to an operator that learns a weight (4.0% for the sample-wise gate and multi-head cross-attention against 0.0% for fixed concatenation, static weighted summation, and the temporal-only control), and it is more frequent on the graph construction that was previously said to remove it. Epoch-level measurement identifies the mechanism: the gate must escape towards the temporal branch to learn at all, and in nine of ten instrumented runs it does, between epochs 4 and 19; the run that never crosses stays weighted towards a spatial branch that carries almost no standalone signal and never leaves $F1 \approx 0.08$. Fifth, the spatial branch cannot be shown to contribute: removing message passing entirely, shuffling the node features, and replacing the topology with random edges each leave performance unchanged within the noise, and under cross-validation the branch shows no significant benefit on any anomaly subtype. The one performance difference this study can resolve runs against the proposed architecture. Gradient boosting on 38 window aggregates reaches 0.9195 ± 0.0307 F1 against

0.8608 ± 0.1464 for the gated hybrid, a trip-level difference of $+0.0409$ (95% CI $[+0.0278, +0.0552]$, Holm $p = 0.0002$), and a zero-parameter three-threshold classifier derived from the generation rules reaches 0.8938 ± 0.0375 —a higher mean than every learned model in the study. Replication across four independently generated datasets spanning two road networks leaves 0 of 40 pairwise comparisons significant. The study argues that architectural claims in this problem setting require multi-partition, multi-seed evidence at the correct statistical unit, and demonstrates on its own earlier results what happens when they do not have it.

Index Terms—Traffic Anomaly Detection, Spatio-Temporal Modeling, Graph Attention Network (GAT), Long Short-Term Memory (LSTM), Multimodal Fusion, Ablation Study, Evaluation Methodology, Clustered Data, Cluster Bootstrap, Permutation Test, Reproducibility, Intelligent Transportation Systems.

I. INTRODUCTION

Intelligent Transportation Systems (ITS) have become a fundamental component of modern urban mobility management by enabling large-scale monitoring and analysis of traffic conditions through heterogeneous data sources. These systems generate massive volumes of spatio-temporal traffic data, providing new opportunities for understanding driving behavior and enhancing road safety. Traffic anomalies—such as sudden stops or abnormal maneuvers—can disrupt traffic flow and create significant safety risks. Accurate and timely detection is therefore critical for decision-support mechanisms in ITS environments.

Recent studies have leveraged deep learning, particularly **Long Short-Term Memory (LSTM)** networks, to model sequential vehicle trajectories. However, road segments form complex graph-structured networks, making purely temporal reasoning insufficient for context-sensitive anomaly detection. To capture structural dependencies, **Graph Attention Networks (GAT)** and related graph-based approaches have been increasingly adopted alongside temporal encoders. As reflected in recent literature—across road-traffic detection and forecasting, generic time-series anomaly detection, and network-traffic monitoring—current research increasingly moves toward both richer spatio-temporal modeling and more effective fusion strategies for integrating temporal behavior with spatial context

[1]–[4]. Existing approaches frequently suffer from “**contextual blindness**,” where a model flags natural behavior as anomalous due to limited structural awareness, while many multimodal methods still rely on static or weakly adaptive fusion that cannot fully respond to changing traffic conditions [2], [4].

A recurring methodological pattern accompanies this progress. Architectural contributions in this area are commonly evaluated through a *single* training run on a *single* train/test partition, and improvements of one to three F1 points are reported as evidence that a proposed fusion or attention mechanism is responsible for the gain. Two properties of the anomaly-detection setting make such evidence fragile. First, anomaly prevalence is low, so the number of positive test instances is small and paired significance tests have limited power. Second, graph message passing on GPU is not bit-reproducible—scatter-based aggregation performs floating-point summation in a nondeterministic order—so repeated runs of an identical configuration do not return identical numbers. Under these conditions, a single-run difference cannot distinguish an architectural effect from seed and partition variance.

This paper therefore asks a narrower and more answerable question than “is the proposed architecture better?”: *which components of a hybrid LSTM–GAT anomaly detector actually carry the observed performance, and how much of the reported advantage survives a controlled protocol?* We keep the task formulation, dataset, grouped partition, optimizer, epoch budget, checkpoint criterion, and threshold-selection rule fixed, vary one component at a time, and report every result over multiple seeds with interval estimates and paired tests. The system under study is the hybrid design we previously proposed, denoted **STANet**, whose fusion stage is a sample-wise scalar modality gate; the evaluation, however, is deliberately symmetric and does not privilege that design.

The main contributions are:

- **Controlled encoder benchmark.** Nine encoder configurations—covering the core operators of STGCN, DCRNN, and Graph WaveNet adapted to the present task—are compared under an identical protocol over five seeds. The comparison separates two questions that are usually reported together, and it separates both from a third: what the design can resolve at all. Graph attention ranks above Chebyshev convolution, diffusion convolution, and adaptive adjacency, but with five seeds on one partition only the gap against adaptive adjacency survives multiplicity correction, and that configuration is unstable rather than merely weaker—its five runs span 0.0203 to 0.7950 F1. Differences along the *temporal* axis are smaller still: the LSTM, a GRU, and a Transformer encoder finish within 0.0015 F1 of one another, an order of magnitude below the leader’s own seed standard deviation, and an equivalence test places one of those three pairs inside a pre-specified bound while leaving the other two undetermined. We report which comparisons are resolved, which are equivalent, and which the data cannot decide, rather than treating the last two as the same outcome.
- **Direct fusion ablation, and a ranking that will not hold still.** Four fusion operators—sample-wise gating, fixed

concatenation, static weighted summation, and multi-head cross-attention—are compared with all other components frozen. The sample-wise gate ranks last of four on the fixed partition, first under cross-validation over two seeds, and fourth over five; the control without a spatial branch ranks first. We report all three orderings rather than one, because the difference between them is the number of seeds averaged and nothing else. Excluding degenerate runs the four operators span 0.0095 F1, under a quarter of a single run’s replication error.

- **A degenerate failure mode attributed to a class of operators.** A reproducible collapse is isolated and attributed by measurement rather than by narrative. Over 500 repeated grouped partitions it occurs in 8 runs, all of them belonging to the two operators that learn a mixing weight and none to the three that do not, and it is *more* frequent on the graph construction an earlier version of this work claimed removed it. Epoch-level probes of the gate value, branch embedding norms, and per-branch gradient norms identify the mechanism.
- **Evidence on evaluation protocol.** We show that an ablation conclusion drawn from one fixed grouped partition is not reproduced under five-fold trip-grouped cross-validation covering all sequences, and that a conclusion drawn from two seeds is not reproduced over five. We further show that the windows used as observations overlap by 80%, so the window-level tests customary in this literature overstate the effective sample size; all inference here uses the trip as the unit, with cluster bootstrap, permutation tests, Holm correction, and the Nadeau–Bengio correction for the training-set overlap that cross-validation induces.
- **Reference points and transparency.** A zero-parameter threshold-rule classifier derived from the generation thresholds, gradient-boosting baselines on window aggregates, probability calibration, and single-window latency on GPU and CPU are reported alongside the learned models, together with a complete data card for the synthetic dataset. Every result file carries a protocol hash, and configurations appearing in several tables are trained once and shared, so the tables are consistent by construction rather than by assertion.

II. RELATED WORK

The literature most relevant to this study covers both anomaly-detection models and the fusion strategies used to integrate temporal and spatial information.

A. Temporal, Spatial, and Spatio-Temporal Anomaly Models

Early traffic anomaly detection studies were predominantly driven by temporal modeling, where recurrent architectures such as LSTM-based networks were used to capture abrupt variations in velocity, acceleration, and short-term driving behavior. These models remain effective when anomalies are primarily reflected in kinematic changes, but they often overlook whether a maneuver is contextually plausible for the surrounding road environment. In parallel, graph-based and

spatially aware methods improved structural understanding by modeling road connectivity, segment interactions, and topological constraints. Such approaches are better suited to representing where risky conditions may emerge, yet they can miss the sequential behavioral evolution that distinguishes a routine slowdown from a truly hazardous event.

Recent studies increasingly bridge these two views through spatio-temporal learning, though not always on the same task. Huang et al. [1] combine Dynamic GAT and Informer components to capture long-range temporal dependencies together with adaptive relational structure, for anomaly detection in generic multivariate time series rather than road traffic. Duan et al. [2] extend this direction through a multimodal deep-fusion framework with heterogeneous graph neural networks. Nguyen et al. [3] further show that graph-based spatio-temporal integration can effectively represent coupled speed–volume patterns in urban traffic environments, in a forecasting rather than a detection setting. Taken together, these studies report that anomaly detection benefits from combining behavioral evolution with structural context; how much weight each modality should receive, however, is usually settled by architectural configuration rather than measured directly.

B. Fusion Strategies and Context Integration

A second research direction focuses on how information from temporal and spatial branches should be fused. Liang et al. [4] adopt a lightweight spatio-temporal architecture that emphasizes compact feature fusion; their setting is *network* traffic in IoT deployments rather than road traffic, and we cite it for the fusion mechanism, not for the task. Cui et al. [5] use a spatial–temporal attention design for lane-aware trajectory prediction. These studies show that attention and feature-level coupling improve representation quality, yet the relative contribution of temporal and spatial cues is still largely determined by architectural configuration rather than by an explicit, empirically validated comparison against simpler fusion operators.

The mechanism examined in this paper is a **sample-wise scalar modality gate** applied after the LSTM and GAT outputs have been projected into a shared 64-dimensional space. It produces one scalar per sample and broadcasts that value across all latent features; it is not a feature-level attention vector, a time-step attention mechanism, or a graph-neighborhood coefficient. Table 1 places this mechanism among related designs. A central empirical question of this study is whether that modality-level distinction translates into a measurable advantage; Section VII-C reports that, under a controlled protocol, it does not.

Beyond fusion design, transfer between simulated and real operating conditions is an open problem for simulation-trained detectors. Digital twin-assisted graph contrastive domain adaptation has shown promise for adapting simulation-derived representations to small-sample target domains in adjacent diagnostic fields [6], and we return to this direction in Section VIII-C.

Table 1. [Literature, as reported by the cited works] Related spatio-temporal architectures and their fusion mechanisms. The works differ in task and domain; only one addresses road-traffic anomaly detection, and the comparison here is over the fusion mechanism rather than over results.

Study	Year	Task and domain	Core Arch.	Temporal	Spatial	Fusion
Huang et al. [1]	2024	Anomaly detection, generic multivariate time series	Dynamic GAT-Informer	Informer	Dynamic GAT	Integrated gated feature fusion with graph-neighborhood attention
Duan et al. [2]	2025	Anomaly detection, highway traffic	Multimodal deep fusion + HGNN	Traffic flow / speed dynamics	Heterogeneous GNN	Integrated multimodal feature fusion
Nguyen et al. [3]	2026	Forecasting (volume, speed), urban road traffic	Spatiotemporal prediction model	Time-series analysis	Graph-based dependency modeling	Integrated spatio-temporal feature fusion
Liang et al. [4]	2025	Anomaly detection, network traffic (IoT)	Lightweight ST network	TCN / CNN	Spatio-temporal features	Compact feature-level fusion
Cui et al. [5]	2026	Trajectory forecasting, road traffic	Lane-flow fusion attention	Temporal self-attention	Lane-level spatial attention	Feature-level attention fusion
This study	2026	Anomaly detection, road traffic (synthetic)	Hybrid LSTM-GAT, evaluated component-wise	Stacked LSTM (vs. GRU, gated TCN, dilated conv, Transformer)	GAT (vs. Chebyshev, diffusion, adaptive adjacency, none)	Four operators compared directly: gated, concat, weighted sum, cross-attention

III. PROBLEM FORMULATION

Urban traffic systems can be modeled as complex spatio-temporal processes in which vehicle movements evolve while simultaneously occurring within the structural constraints of a road network [7], [8]. In this study, the traffic anomaly detection problem is formulated as a context-aware learning task that jointly models vehicle trajectory dynamics and road network topology [9]. The mathematical notations used throughout this formulation are summarized in Table 2.

A. Road Network Representation

The road network is represented as a directed graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$, where $\mathcal{V}$ is a set of N nodes (road segments) and $\mathcal{E}$ denotes the set of edges representing physical connectivity [10], [11]. Each node $v_i \in \mathcal{V}$ is associated with a structural feature vector $\mathbf{s}_i$:

$$\mathbf{s}_i = [\text{type}_i, \text{length}_i, \text{deg}_i, \text{context}_i] \quad (1)$$

where type_i denotes the road category, length_i is the segment length, deg_i is the node degree, and context_i represents additional intersection characteristics [12], [13]. Spatial dependencies are encoded via an adjacency matrix $\mathbf{A}$ [14], [15].

B. Vehicle Trajectory Modeling

Vehicle movement is represented as a trajectory sequence $\mathcal{T} = \{\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_\tau\}$, where τ is the sequence length and each observation $\mathbf{x}_t$ describes the kinematic state at time t [16], [17]:

$$\mathbf{x}_t = [v_t, a_t, \theta_t, id_t] \quad (2)$$

where v_t is speed, a_t is acceleration, θ_t is the heading angle, and id_t is the identifier of the road segment where the vehicle is located.

C. Spatio-Temporal Feature Representation

Anomaly detection requires the integration of behavioral dynamics and structural context [18], [19]. We define the learning task as finding a joint embedding $\mathbf{z}$:

$$\mathbf{z} = f(\mathbf{h}_{temp}, \mathbf{h}_{spat}) \quad (3)$$

where $\mathbf{h}_{temp} = \text{LSTM}(\mathcal{T})$ and $\mathbf{h}_{spat} = \text{GAT}(\mathcal{G}, \mathbf{A})$ [20], [21]. The function $f(\cdot)$ denotes the fusion operator, which is treated in this study as an *experimental variable* rather than a fixed design choice.

D. Traffic Anomaly Detection Objective

The detection task is formulated as a binary classification:

$$\hat{y} = \sigma(g(\mathbf{z})) \quad (4)$$

where $g(\cdot)$ is the classification head and $\hat{y} \in [0, 1]$. At the task-formulation level:

$$\mathcal{L} = -\frac{1}{M} \sum_{i=1}^M [y_i \log(\hat{y}_i) + (1 - y_i) \log(1 - \hat{y}_i)] \quad (5)$$

where $y_i \in \{0, 1\}$ is the ground truth label and M is the number of samples [2], [22], [23]. Equation 5 is the generic task-level formulation only; because the training partition is imbalanced, the implemented objective is the weighted binary cross-entropy in Eq. 20.

E. Problem Definition

Given a road network $\mathcal{G}$, vehicle trajectory $\mathcal{T}$, and road features $\mathcal{S}$, the objective is to learn a mapping $\Phi : \{\mathcal{T}, \mathcal{G}, \mathcal{S}\} \rightarrow \{0, 1\}$ [9], [24].

Table 2. Summary of mathematical notation

Symbol	Description
$\mathcal{G} = (\mathcal{V}, \mathcal{E})$	Directed graph representing the road network
$\mathcal{V}, \mathcal{E}$	Set of nodes (road segments) and edges (connectivity)
$N, \mathbf{A}$	Number of nodes and adjacency matrix encoding connectivity
$\mathbf{s}_i, \mathcal{S}$	Structural feature vector of node i and the set of all structural features
$\mathcal{T} = \{\mathbf{x}_1, \dots, \mathbf{x}_\tau\}$	Vehicle trajectory sequence of length τ
$\mathbf{x}_t, \mathbf{X} \in \mathbb{R}^{\tau \times 4}$	Per-step kinematic vector and stacked temporal feature matrix
v_t, a_t, θ_t, id_t	Velocity, acceleration, heading angle, and road-segment ID at time t
x_{sin}, x_{cos}	Trigonometric heading components used for continuous angular representation
$type_i, length_i$	Road category and segment length;
$deg_i, context_i$	node degree and contextual attributes
$\mathbf{h}_t, \mathbf{h}_{temp}, \mathbf{h}_{spat}$	LSTM hidden state, temporal embedding, and spatial embedding
$\Theta_{temp}, \Theta_{spat}$	Learnable parameter sets of the LSTM and GAT branches
$\alpha_{ij}, \mathcal{N}_i$	Graph-attention coefficient from node j to node i and neighborhood of node i
$\mathbf{W}, \vec{a}, \mathbf{h}'_i$	GAT transform matrix, attention vector, and updated node representation
d	Shared latent dimension used before multi-modal fusion
$\mathbf{h}'_{temp}, \mathbf{h}'_{spat}$	Projected temporal and spatial embeddings in shared latent space
$\mathbf{W}_t, \mathbf{b}_t$	Learnable projection parameters for temporal/spatial branches
$\mathbf{W}_s, \mathbf{b}_s$	
$g_{gate}, \mathbf{W}_g, b_g$	Sample-wise scalar modality weight and gating-network parameters
$\mathbf{z}$	Joint spatio-temporal fused embedding
$f_{LSTM}(\cdot), f_{GAT}(\cdot)$	Temporal encoder, spatial encoder,
$f_{fusion}(\cdot), g(\cdot)$	fusion, and classification mappings
$\sigma(\cdot)$	Activation function (sigmoid where specified)
$y, \hat{y}$	Ground-truth and predicted labels (global and sample-level forms)
$y_i, \hat{y}_i$	
$\mathcal{L}, M$	Loss function and mini-batch size
w_{pos}	Positive-class weight in the WBCE loss
Φ	Overall mapping function for anomaly detection

IV. ARCHITECTURE UNDER STUDY

This section describes the hybrid architecture whose components are evaluated in Section VII. The model jointly encodes temporal vehicle dynamics and spatial road-network topology through parallel branches and combines them with a configurable fusion operator. We describe the mechanism here and defer all claims about its effectiveness to the experimental sections; the protocol governing those experiments is given in Section VI.

A. Overall Architecture

As illustrated in Fig. 1, the framework consists of (i) a temporal module for sequence modeling, (ii) a spatial module for structural graph learning, and (iii) a fusion module that combines the two projected representations.

Given a vehicle trajectory sequence $\mathcal{T}$ and the road network graph $\mathcal{G}$, the model learns two complementary latent embeddings:

$$\mathbf{h}_{temp} = f_{LSTM}(\mathcal{T}; \Theta_{temp}) \quad (6)$$

$$\mathbf{h}_{spat} = f_{GAT}(\mathcal{G}, \mathbf{A}; \Theta_{spat}) \quad (7)$$

where Θ_{temp} and Θ_{spat} denote the learnable parameter sets of the two branches [21], [25]. These embeddings are integrated through the fusion operator:

$$\mathbf{z} = f_{fusion}(\mathbf{h}_{temp}, \mathbf{h}_{spat}) \quad (8)$$

The fused representation $\mathbf{z}$ is then mapped by a classification head to the anomaly probability $\hat{y}$.

B. Temporal Module

1) *Kinematic Feature Representation*: The temporal branch processes four kinematic features at each time step t :

$$\mathbf{x}_t = [v_t, a_t, \sin(\theta_t), \cos(\theta_t)] \quad (9)$$

Representing the heading angle θ_t using its trigonometric components prevents periodic discontinuities at the angular boundaries [17]. Each driving window is represented as $\mathbf{X} \in \mathbb{R}^{\tau \times 4}$ with $\tau = 50$.

2) *LSTM-Based Sequence Encoding*: A two-layer stacked LSTM encoder is used [16], [17], [26]:

$$\mathbf{h}_t = \text{LSTM}(\mathbf{x}_t, \mathbf{h}_{t-1}) \quad (10)$$

Each layer contains 128 hidden units with dropout 0.2 between recurrent layers. The final hidden state of the second layer is exported as the temporal embedding, $\mathbf{h}_{temp} = \mathbf{h}_\tau$. Table 3 summarizes the branch. Section VII-B replaces this encoder with four alternatives while keeping everything else fixed.

Table 3. Summary of LSTM branch configuration

Component	Setting	Role / Output
Input window	$\tau = 50$, 4 features	Tensor shape $[B, 50, 4]$
Input features	$v_t, a_t, x_{sin}, x_{cos}$	Driving-profile representation
Stacked LSTM	input size = 4, hidden size = 128	Temporal dependency learning
Number of layers	2	Hierarchical sequence encoding
Dropout	$p = 0.2$	Regularization between LSTM layers
Temporal embedding	$\mathbf{h}_{temp} \in \mathbb{R}^{128}$	Output tensor $[B, 128]$
Parameters	200,704	89.2% of the full model

C. Spatial Module

1) *Graph-Based Contextual Features*: The spatial module uses a GAT to encode the road network structure [7], [10], [13]. Each node v_i is associated with a 17-dimensional structural vector $\mathbf{s}_i$ capturing road-type descriptors, topological metrics (betweenness and closeness centrality), and contextual constraints (traffic-light indicators, tunnel information, rarity scores).

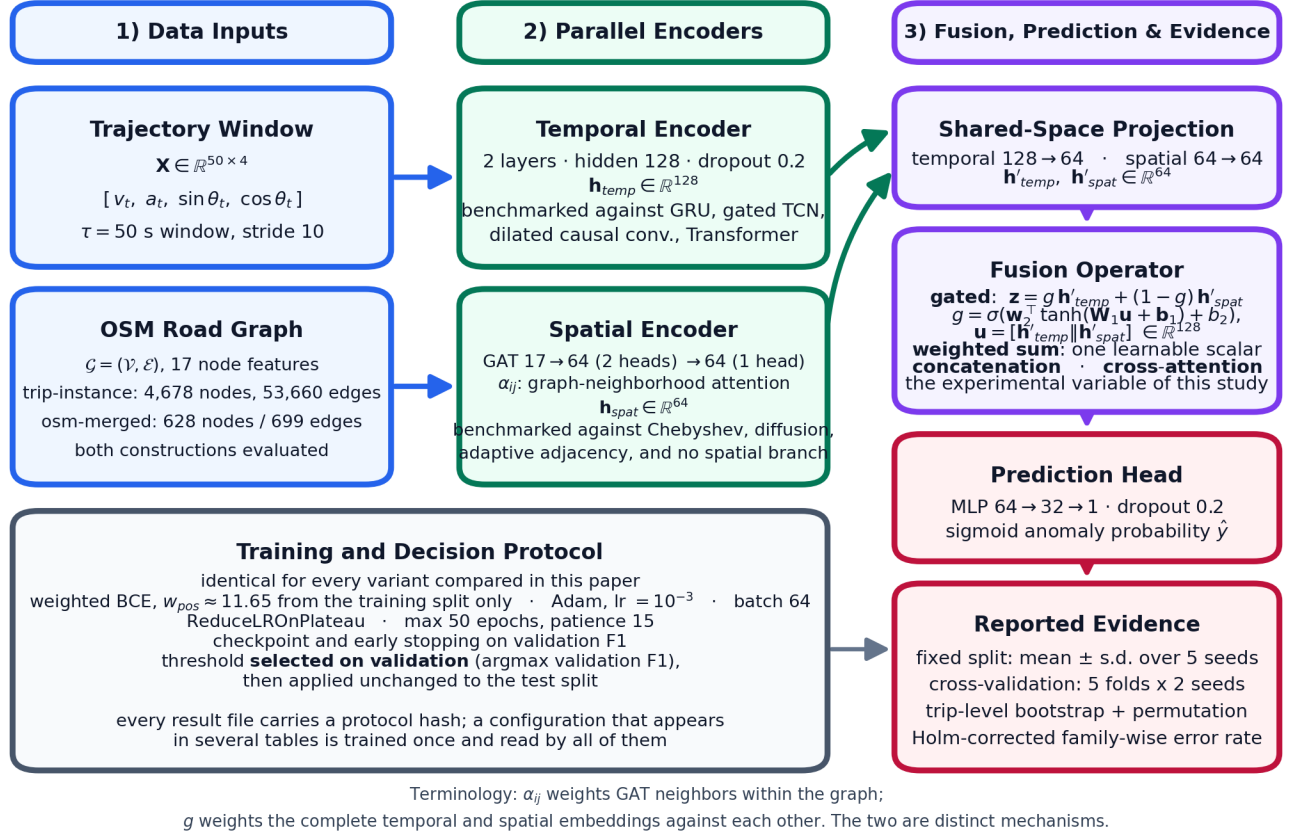


Figure 1. Overall architecture. The GAT neighborhood coefficients α_{ij} are distinct from the sample-wise scalar modality gate g_{gate} ; the fusion block is the experimental variable of Section VII-C.

2) *Graph Attention Mechanism*: The GAT layers compute adaptive attention coefficients α_{ij} over adjacent road segments [9], [20], [27]:

$$\alpha_{ij} = \frac{\exp(\text{LeakyReLU}(\bar{a}^T [\mathbf{W}\mathbf{s}_i \parallel \mathbf{W}\mathbf{s}_j]))}{\sum_{k \in \mathcal{N}_i} \exp(\text{LeakyReLU}(\bar{a}^T [\mathbf{W}\mathbf{s}_i \parallel \mathbf{W}\mathbf{s}_k]))} \quad (11)$$

$$\mathbf{h}'_i = \sigma \left(\sum_{j \in \mathcal{N}_i} \alpha_{ij} \mathbf{W}\mathbf{s}_j \right) \quad (12)$$

Two consecutive GAT layers are used: the first maps the 17-dimensional node attributes to a 64-dimensional intermediate representation (32 hidden channels, 2 attention heads); the second produces the final 64-dimensional spatial embedding. The embedding of the road segment occupied by the vehicle is extracted as $\mathbf{h}_{spat}$. Table 4 summarizes the branch.

Table 4. Summary of GAT branch configuration

Component	Setting	Role / Output
Graph input	4,678 nodes, 53,660 edges	Trip-specific segment instances
Distinct physical roads	628 unique <code>osm_segment_id</code>	See Section V-B
Node features	17 dimensions	Structural and contextual descriptors
GAT layer 1	GATConv(17, 32, heads=2)	Intermediate tensor $[N, 64]$
GAT layer 2	GATConv(64, 64, heads=1)	Spatial embedding tensor $[N, 64]$
Spatial embedding	$\mathbf{h}_{spat} \in \mathbb{R}^{64}$	Output of active road segment
Parameters	5,568	2.5% of the full model

D. Fusion Module

Since $\mathbf{h}_{temp}$ and $\mathbf{h}_{spat}$ originate from different branches, they are first projected into a common latent dimensionality $d = 64$:

$$\mathbf{h}'_{temp} = \text{ReLU}(\mathbf{W}_t \mathbf{h}_{temp} + \mathbf{b}_t) \quad (13)$$

$$\mathbf{h}'_{spat} = \text{ReLU}(\mathbf{W}_s \mathbf{h}_{spat} + \mathbf{b}_s) \quad (14)$$

Four fusion operators are considered. The **sample-wise scalar modality gate** originally proposed computes Eq. 15 and applies it in Eq. 16. The gate is a two-layer network, not the single projection written in the previous version of this paper: $\mathbf{W}_1 \in \mathbb{R}^{32 \times 128}$ and $\mathbf{w}_2 \in \mathbb{R}^{32}$, giving the 4,161 parameters listed in Table 5. A reviewer noted the discrepancy between the equation and the table; the table and the released code were correct.

$$g_{gate} = \sigma(\mathbf{w}_2^T \tanh(\mathbf{W}_1 [\mathbf{h}'_{temp} \parallel \mathbf{h}'_{spat}] + \mathbf{b}_1) + b_2) \quad (15)$$

$$\mathbf{z} = g_{gate} \cdot \mathbf{h}'_{temp} + (1 - g_{gate}) \cdot \mathbf{h}'_{spat} \quad (16)$$

where $g_{gate} \in [0, 1]$ has tensor shape $[B, 1]$: exactly one scalar per sample, broadcast across all 64 dimensions. The three comparison operators are **fixed concatenation** ($\mathbf{z} = [\mathbf{h}'_{temp} \parallel \mathbf{h}'_{spat}]$, no reweighting), **static weighted summation** ($\mathbf{z} = s \mathbf{h}'_{temp} + (1 - s) \mathbf{h}'_{spat}$ with a single global learnable scalar s), and **multi-head cross-attention** (four heads, temporal embedding as query, both embeddings as key/value).

We note a structural relation that motivates the ablation. If g_{gate} converges to a constant $\bar{g}$, then Eq. 16 followed by a linear classifier $\mathbf{W}$ yields $\mathbf{W}\mathbf{z} = \bar{g}\mathbf{W}\mathbf{h}'_{temp} + (1 - \bar{g})\mathbf{W}\mathbf{h}'_{spat}$, which fixed concatenation can reproduce exactly by setting its two weight blocks to $\bar{g}\mathbf{W}$ and $(1 - \bar{g})\mathbf{W}$. Concatenation therefore spans a strictly larger function class than constant-gate fusion, and any advantage of gating must come from genuinely sample-dependent behavior. Section VII-D tests whether such behavior is observed.

The fused representation passes through a two-layer MLP ($64 \rightarrow 32 \rightarrow 1$, ReLU, dropout 0.2) to produce the anomaly probability. Table 5 summarizes the stage.

Table 5. Summary of the fusion stage and the four operators compared

Component	Setting	Fusion parameters
Temporal projection	Linear $128 \rightarrow 64 + \text{ReLU}$	—
Spatial projection	Linear $64 \rightarrow 64 + \text{ReLU}$	—
Gated (proposed)	Linear $128 \rightarrow 32 + \tanh$, $32 \rightarrow 1 + \text{sigmoid}$	4,161
Fixed concatenation	$[\mathbf{h}'_{temp} \parallel \mathbf{h}'_{spat}]$	0
Static weighted sum	single learnable scalar s	1
Cross-attention	4-head, $d = 64$	16,640
Classifier	MLP $64/128 \rightarrow 32 \rightarrow 1$	—

V. DATASET AND PREPROCESSING

A. Data Sources

The empirical evaluation relies on a **physics-informed synthetic vehicle trajectory dataset** and **OpenStreetMap (OSM)** road network topology.

1) *Vehicle Trajectory Dataset*: **No pre-existing real-world source dataset was used.** To obtain trajectory data with fully known ground-truth anomaly labels—difficult to obtain at scale from real sensor logs because of privacy constraints, sparse anomaly incidence, and the absence of verified incident annotations—this study employs a physics-informed synthetic trajectory-generation framework. Vehicle motion is simulated over the OSM-derived Kocaeli road network using a Markov-chain behavioral model in which each vehicle transitions between four kinematic states (constant speed, acceleration, deceleration, stop) at one-second resolution. State-transition probabilities are conditioned on the local traffic regime (free-flow, transition, congested), defined through a log-normal traffic-intensity distribution and a speed-density relationship adapted from Pipes’ car-following model [34]:

$$\bar{v}_s = (v_{\max} - 0.08v_{\max}) \left(1 - \frac{\tau}{10}\right)^4, \quad (17)$$

where $v_{\max}$ is the segment’s speed limit and $\tau \in [1, 9]$ is the sampled traffic-intensity level. Acceleration is constrained to $[-3, 1.5]$ m/s² under normal driving, following AASHTO geometric-design guidance [35]. Sensor-level noise is injected into the speed and bearing channels at every time step.

The generated dataset contains 293,168 records across 90 inter-district trips. Ground-truth labels are recorded at injection time according to the taxonomy in Table 8. A complete data card—generation parameters, subtype rules, imputation shares, and known limitations—is released with the code and data (Section VIII-C).

2) *OpenStreetMap Road Network Data*: Each road segment is represented by a 17-dimensional feature vector $\mathbf{s}_i$ [13] comprising structural indicators (lane count, road category), infrastructure elements (traffic lights, tunnels), and topological measures (betweenness and closeness centrality).

Several structural attributes required imputation because of incomplete OSM tagging. Lane counts and speed limits, when missing, were assigned the road-type-conditional defaults in Table 6; assigned speed values were additionally smoothed using

$$\hat{v}_i = \begin{cases} \text{Round}_5\left(\frac{v_{i-1} + v_i + v_{i+1}}{3}\right), & C_i, \\ v_i, & \text{otherwise,} \end{cases} \quad (18)$$

$$C_i \equiv (S_i \neq \text{OSM}) \wedge (|v_{i-1} - v_i| > 30),$$

to remove physically implausible discontinuities. Traffic-light presence was inferred using a hybrid rule combining OSM `traffic_signals` tags with a node-degree heuristic.

The values in Table 6 are **author-defined imputation assumptions**, not observed OSM attributes or statutory limits. **53.8% of lane counts and 35.3% of speed limits originate from OSM**; the remainder are imputed defaults (46.2%) or smoothed values (14.8% of speeds). This is material for interpreting the spatial branch, because $v_{\max}$ is one of its node features.

Table 6. Author-defined road-type-conditional imputation defaults

Road Type	Default Lanes	Default Speed (km/h)
motorway	3	130
motorway_link	1	80
trunk	2	110
trunk_link	1	60
primary	2	90
primary_link	1	50
secondary	1	80
secondary_link	1	40
(default)	—	50

The network covers 10 central districts of Kocaeli, Türkiye, with 90 inter-district routes forming the trip set.

B. Graph Construction and Node Multiplicity

The graph is constructed by treating each *trip-specific traversal* of a road segment as a separate node. This yields 4,678 nodes and 53,660 directed edges, with an average node degree of 11.47. It is important to state the structure this reflects: the 4,678 nodes correspond to only **628 distinct physical roads** identified by `osm_segment_id`, so each physical road appears on average 7.4 times, once per trip that traverses it. Node features are identical across the copies of a given physical road in 538 of 546 multi-copy groups.

Merging the copies into a single node per physical road yields a substantially different graph: 628 nodes, 699 edges, and an average degree of 1.11. The reported degree of 11.47 therefore describes the trip-instance graph, not the underlying road network, which is closer to a set of overlapping route corridors than to a densely connected urban network.

Either construction imposes a ceiling on what the spatial branch alone can achieve. Because several sequences with different labels map to the same node, road identity alone bounds achievable accuracy at 94.58% on the trip-instance graph and 92.84% on the merged graph. Both figures are computed from **training labels only**; the reason is given below. The spatial branch can therefore act as a context provider but cannot serve as a standalone classifier—a prediction confirmed in Section VII-B.

a) Neither construction is selected; both are experimental factors: The previous version of this work designated the trip-instance graph as the primary setting and justified that designation by its higher per-node label purity. Two problems with that justification are worth stating, because the second is the more serious.

The first is that an earlier version justified the same choice by downstream performance, which makes the graph a function of the outcome it is used to support. Replacing that with label purity was intended to fix it.

The second is that the replacement was worse. The purity ceiling was computed over the training, validation *and* test partitions, so a model-selection criterion was reading test labels. A reviewer identified this, and the objection is correct.

We therefore do not select a graph at all. Both constructions are treated as **prespecified experimental factors**, and every result in this paper is reported under both, at the same seeds and the same partitions (Section VII-O). Where the two disagree we say so rather than resolving the disagreement in favour of either. This removes the need for a selection criterion, and with it the opportunity for one to leak.

The label-purity figures quoted above are now computed from training labels alone and are reported as a descriptive property of the data, not as grounds for a choice. The structural trade-off remains worth stating: the merged graph is the faithful representation of the physical road network, while the trip-instance graph distributes sequences over more nodes and raises per-node label purity at the cost of duplicating each road once per traversal, which inflates the average degree and weakens message passing between genuinely adjacent roads.

1) Recovering the Transition Behaviour: The generator's state-transition probabilities were not available to us. What can be documented is the behaviour they produced, and that is recoverable from the released trajectories. Each one-second step of every non-anomalous record is assigned to one of the four states—anomalous records are excluded because an anomaly scenario overrides the Markov process—using $v < 1.0$ km/h for *stop* and $|a| > 0.10$ m/s² to separate *acceleration* and *deceleration* from *constant speed*. Transitions are counted only between consecutive seconds within a trip.

Two properties are worth stating because neither is visible from the model description. First, the regimes behave as intended: mean speed falls from 57.8 to 25.0 km/h between free-flow and congested conditions, and the share of steps spent stopped rises from 0.0001 to 0.0565.

Second, the *constant speed* state is almost unused. Across all regimes it accounts for 0.0103 of steps, against 0.6464 accelerating and 0.3368 decelerating. This is not an artefact of the threshold: widening it fivefold, from 0.05 to 0.50 m/s²,

raises the free-flow share only from 0.0072 to 0.0632. The simulated vehicle is in near-continuous alternation between accelerating and decelerating: from a decelerating step in free-flow, 0.968 of transitions go to an accelerating one. A four-state description is therefore generous; the realised process is close to two-state with an absorbing stop in congestion.

What this is and is not. These are the transition frequencies the released data exhibits, not the parameters the simulator was given. Three things separate them: anomaly scenarios suspend the Markov process and are excluded here, speed limits and acceleration bounds constrain which transitions are reachable, and sensor noise enters the state assignment through the acceleration channel. We report them because reproducibility requires knowing how the data behaves, and because a reader can check them against any regenerated dataset. How the underlying probabilities were chosen or calibrated remains undocumented, and we record that as a limitation rather than infer it.

C. Data Cleaning and Feature Engineering

Records with implausible sensor readings were removed before sequence construction. Of 293,168 generated records, 60 carry a speed outside $[0, 200]$ km/h and 8,186 a road gradient outside $[-50\%, 50\%]$; one record violates both, so the union is 8,245 records (2.81%), leaving 284,923:

$$\text{retain row } i \iff 0 \leq v_i \leq 200 \wedge -50 \leq r_i \leq 50. \quad (19)$$

The raw heading angle is transformed into sine and cosine components to prevent discontinuities at the $0^\circ/360^\circ$ boundary [17].

D. Anomaly Taxonomy and Label Generation

Anomalies are injected during simulation according to the two-category, nine-subtype taxonomy in Table 8. Of the records the simulator produces, 6.86% carry a non-zero anomaly label; after cleaning this becomes 7.01%, and after windowing 7.07% of sequences are anomalous. The three figures are quoted at three different stages of the pipeline and are not interchangeable.

1) A Conflict Between the Generation and Cleaning Rules: Auditing the two rule sets against each other exposes a problem that the aggregate counts hide. Subtype 6, the GPS speed artifact, is produced by perturbing the speed channel to an implausible value; the cleaning rule discards records whose speed is implausible. The two rules act on the same records, and the cleaning rule wins: of the 60 records removed by the speed filter, 55 are subtype 6, and the subtype falls from 66 generated records to 11 surviving ones. Their speeds span 95.6 to 462.1 km/h, so every instance above 200 km/h is deleted by construction.

This is not a rounding detail. It means the evaluation set contains almost none of an anomaly class the generator was designed to create, and it explains why subtype 6 carries so little weight in the per-subtype results of Section VII-E1: there is nearly nothing left to detect. We report the subtype rather than dropping it, and we state that its numbers describe eleven records. A pipeline intended to evaluate detection of implausible

Table 7. **[Record level, non-anomalous steps only]** Transition probabilities recovered from the released data, by traffic regime. Rows are the current state, columns the next. The final column is the share of steps spent in each state.

Regime	From	Stop	Decel.	Const.	Accel.	Occupancy
free-flow	Stop	0.000	0.000	0.000	1.000	0.0001
	Decel.	0.000	0.029	0.003	0.968	0.3367
	Const.	0.000	0.076	0.645	0.279	0.0126
	Accel.	0.000	0.502	0.006	0.492	0.6507
transition	Stop	1.000	0.000	0.000	0.000	0.0029
	Decel.	0.000	0.047	0.002	0.951	0.3283
	Const.	0.000	0.333	0.468	0.199	0.0045
	Accel.	0.000	0.460	0.003	0.537	0.6643
congested	Stop	0.961	0.000	0.000	0.039	0.0565
	Decel.	0.008	0.171	0.001	0.819	0.3518
	Const.	0.000	0.664	0.204	0.133	0.0039
	Accel.	0.000	0.473	0.005	0.522	0.5878

Table 8. **[Record level, before and after cleaning]** Anomaly taxonomy, generation rules, and observed counts. The two count columns differ because the cleaning rules of Section V-D1 remove anomalous records, most severely for subtype 6.

No.	Subtype	Generation Rule	Gen.	Clean
1	Speed-limit violation	$v_t > 1.10 v_{\max}$	1,405	1,353
2	Abnormal dwell	Stationary for > 30 s	11,610	11,610
3	Abnormal acceleration	$a_t \in [-6, -3] \cup [1.5, 2.5]$ m/s ²	299	287
4	Accident	Composite perturbation + prolonged stop	5,741	5,740
5	Abnormal heading change	$ \Delta\theta_t > 10^\circ$	213	208
6	GPS speed artifact	Perturbation from kinematic history	66	11
7	GPS acceleration artifact	Perturbation from kinematic history	76	75
8	GPS heading artifact	Perturbation from kinematic history	94	93
9	Signal freeze	Identical values for ≥ 2 s	606	593

sensor readings should not filter implausible sensor readings beforehand; correcting this requires regenerating the dataset with the two rule sets reconciled, which is outside the scope of this revision and is recorded as a limitation.

The generation rules are necessary but not always sufficient. A second audit compared each printed rule against the labels actually assigned. For subtype 1 the rule holds in one direction only: every record labelled speed-limit violation satisfies $v_t > 1.10 v_{\max}$, but of the 1,400 cleaned records that satisfy it, 1,353 carry that label, 9 carry subtype 6 instead—a GPS artifact takes precedence—and 38 carry no anomaly label at all. The rules in Table 8 are therefore the conditions under which a subtype *may* be injected, not a complete specification of the labelling function; the remaining condition was not recoverable from the released data. Readers reproducing the labels from the table alone will obtain 3.5% more positives for subtype 1 than the dataset contains.

Label-input dependence. Subtypes 1, 2, 3, 5, and 9 are threshold functions of the same kinematic channels supplied to the temporal branch. This is a property of the dataset, not an artifact of the models, and it is quantified directly in Section VII-A through a zero-parameter rule classifier. One subtype is an informative exception: speed-limit violation depends on $v_{\max}$, which is *not* an input to the temporal branch and is available only as a graph node feature. Section VII-E1 tests whether the spatial branch exploits this.

E. Feature Normalization, Sequences, and Partitioning

Velocity is recorded in km/h at one-second resolution, and acceleration is not an independent channel: it is the first difference of velocity, $a_t = (v_t - v_{t-1})/3.6$ m/s². We state this because the two kinematic inputs are therefore deterministically related, which constrains what an input perturbation can mean—noise added to velocity must be propagated into acceleration, or the network is shown a combination that the data-generating process cannot produce. The relation was verified rather than assumed: over the entire test partition the ratio $(v_t - v_{t-1})/a_t$ equals 3.6000 at both the 5th and the 95th percentile. Velocity is normalized with Min-Max scaling and acceleration with Z-score standardization. **Scalers are fitted on the training partition only** and applied to validation and test data, avoiding the distributional leakage that arises when scaling precedes partitioning.

Trajectories are transformed into fixed-length sequences using a sliding window of $\tau = 50$ steps with a stride of 10, yielding 28,090 sequences of which 7.07% are anomalous. The window label and road segment are taken from the final time step.

As summarized in Section V-E, the dataset is partitioned by unique trip identifiers rather than by individual samples, so that overlapping windows from the same route cannot appear in more than one partition. Two evaluation regimes are used:

- a **fixed grouped partition** (70/15/15 by trip) used for the encoder and fusion comparisons, summarized in Table 9;
- **five-fold trip-grouped cross-validation**, in which every one of the 28,090 sequences receives an out-of-fold test prediction, used in Section VII-E.

Table 9. Fixed grouped partition and anomaly distribution

Partition	Trips	Sequences	Anomaly Ratio
Train	63	19,255	7.90%
Validation	13	4,036	7.58%
Test	14	4,799	3.27%

The lower test prevalence (3.27% versus 7.90% in training) arises from the anomaly composition of the selected whole trips and was not corrected by moving windows across partitions, which would violate the grouping constraint. This imbalance is one motivation for the cross-validated regime, where fold-wise test prevalence ranges from 3.40% to 10.49%.

VI. EXPERIMENTAL PROTOCOL

A. Controlled Comparison Design

Every comparison in this paper varies **one component at a time** while holding the following fixed: task formulation, dataset, grouped partition, optimizer (Adam, learning rate 0.001), batch size (64), maximum epochs (50), early-stopping patience (15), learning-rate schedule (*ReduceLROnPlateau*, factor 0.5, patience 3), gradient clipping (max-norm 1.0), loss function, checkpoint-selection criterion, and threshold-selection rule. Encoder comparisons additionally hold the fusion operator fixed, and fusion comparisons hold both encoders fixed.

1) *Loss Function*: Because anomalies are rare, a **weighted binary cross-entropy** objective is used:

$$\mathcal{L} = -\frac{1}{M} \sum_{i=1}^M [w_{pos} \cdot y_i \log(\hat{y}_i) + (1 - y_i) \log(1 - \hat{y}_i)] \quad (20)$$

with $w_{pos} = N_{neg}^{train} / N_{pos}^{train} \approx 11.65$, computed exclusively from the training partition after the grouped split.

2) *Checkpoint and Threshold Selection*: A single criterion—best validation F1-score—governs both checkpoint retention and early stopping for *all* configurations. Mixing criteria across variants (for example, early-stopping on loss while checkpointing on F1) makes runs incomparable and is avoided.

The decision threshold is selected on the **validation** partition by maximizing validation F1 over a grid, and then applied unchanged to the test partition. Selecting a threshold on held-out validation data is standard practice and does not constitute test leakage; fixing it a priori at 0.50, as in the previous version of this work, discards performance without any methodological benefit, because a model trained with $w_{pos} = 11.65$ is not calibrated to that operating point.

3) *Repeatability and Reporting*: Every learned configuration on the fixed partition is trained with the same five random seeds (42–46), and all results are reported as **mean $\pm$ standard deviation**. The cross-validated analysis uses five trip-grouped folds crossed with the same five seeds, giving twenty-five observations per configuration; an earlier version of this analysis used two, and Section efsec:cv reports what changed when it did not.

Multi-seed reporting is not optional in this setting. Graph message passing uses scatter-based aggregation, whose floating-point summation order is nondeterministic on GPU; enabling deterministic cuDNN algorithms does not remove this nondeterminism, and we measured two runs of an identical configuration and seed differing by 0.025 F1. Bit-level reproducibility is therefore unattainable, and a single reported number carries an unquantified error bar.

4) *Protocol Stamping and Shared Runs*: Because bit-level reproducibility is unattainable, the burden falls on *procedural* consistency: any two numbers presented as comparable must have been produced under exactly the same protocol. An earlier version of this study did not enforce that, and it failed in a way worth reporting. Halfway through the experiment campaign the cuDNN determinism flag was switched off—deterministic algorithms were making the dilated convolution backward pass roughly forty times slower while providing no reproducibility guarantee, for the reason just given—and the experiments run before and after that change were subsequently placed in the same tables. The configuration LSTM + GAT + weighted sum consequently appeared with two different scores in two tables of the same manuscript, with nothing in the record to explain the discrepancy.

Two mechanisms now prevent this. First, each experiment records a *protocol stamp*: a hash over the seed set, epoch budget, patience, batch size, learning rate, scheduler parameters, dropout, checkpoint criterion, threshold rule, cuDNN flags, window length, stride, split seed, and library versions. Scripts that import another experiment’s results assert equality of stamps and abort on mismatch, so results from different protocols cannot silently enter one table. Second, a configuration that appears in more than one table is trained *once* and cached under its protocol stamp; the tables then read the same run. In this paper LSTM + GAT + weighted sum appears in four tables, and all four report the identical cached run rather than four independently trained runs presumed to agree. Changing any protocol field changes the hash, invalidates the cache, and forces retraining.

B. Unit of Statistical Analysis

Windows are extracted with length $\tau = 50$ and stride 10, so two consecutive windows share 40 of their 50 time steps—an overlap of 80%—and all windows drawn from one trip share a driver, a vehicle, and a route. Windows are therefore *not* independent observations, and paired tests that treat them as such (window-level McNemar, window-level bootstrap) overstate the effective sample size: confidence intervals come out too narrow and p -values too small. This is a property of the sampling design, not of any particular model, and it applies to every comparison in this paper.

All inferential statements below therefore use the **trip** as the unit:

- **Cluster bootstrap**. Trips, not windows, are resampled with replacement; a selected trip contributes all of its windows. Confidence intervals are the 2.5th and 97.5th percentiles of 5,000 resampled F1 differences.
- **Cluster permutation test**. Under the null hypothesis that two models are exchangeable within a trip, an independent

coin flip per trip swaps the two models’ predictions on all windows of that trip. The two-sided p -value is the proportion of 20,000 such relabelings whose F1 difference is at least as extreme as the observed one. This is the valid counterpart of McNemar’s test for clustered data.

- **Multiplicity.** Within each family of comparisons (base-lines, fusion operators, encoder configurations) Holm–Bonferroni step-down correction controls the family-wise error rate at $\alpha = 0.05$.

Both quantities are computable in closed form over per-trip contingency counts, since F1 depends only on the totals TP, FP, and FN and these are additive over trips; the implementation was verified against a naive resampling loop.

The consequence is a real loss of statistical power, and we state it plainly rather than absorbing it silently. The dataset contains 90 trips in total, so the fixed test partition contains roughly 13 independent units. Interval estimates on that partition are correspondingly wide. Under cross-validation every one of the 90 trips receives an out-of-fold prediction, which is why the cross-validated results are treated as the primary evidence in this paper and the fixed-partition results as a secondary, more variable view of the same question. Window-level values are additionally reported in the accompanying result files, labelled as such, so that readers can see the magnitude of the difference the correct unit makes.

C. How Much a Single Run Is Worth

The claim that graph message passing is not bit-reproducible on GPU has been made twice above. It is stated in the literature, but this paper relies on it heavily enough that it should be measured here rather than cited, because the size of the effect decides whether any single-run comparison in this area can carry an architectural conclusion.

The measurement is a controlled replication. Two configurations were trained a second time under the same protocol stamp, at the same seed, on the same five cross-validation partitions. Everything that the experimenter controls is identical between the two runs; the only difference is the order in which the GPU accumulates floating-point sums. The two configurations differ in exactly one respect: whether the spatial branch contains a GATConv layer.

Table 10. **[Fixed protocol, same seed, same partition]** Replication of an identical run. Ten pairs, five per configuration.

Configuration	GATConv	Identical	Mean $ \Delta F1 $	Max $ \Delta F1 $
Temporal only	no	5/5	0.0000	0.0000
LSTM + GAT	yes	0/5	0.0195	0.0414

Without the graph layer, replication is exact: five of five pairs agree to every reported digit, and the selected epoch is the same in each. With it, none of the five pairs agree; the mean absolute difference is 0.0195 F1 and the largest is 0.0414. The instability is not confined to the final metric. In one pair the best validation F1 was reached at epoch 4 in the first run and epoch 43 in the second, so the two runs are not small perturbations of one another but different trajectories with similar endpoints.

The number to compare this against is the spread of the architectures themselves. Across the five configurations of the cross-validated fusion comparison (Table 17), the gap between the best and the worst mean F1 is 0.0582 over all runs, but three of the 125 runs are degenerate and carry that gap by themselves; among runs that trained successfully the gap is 0.0095. **A single run’s replication error with graph attention is therefore four times the entire architectural spread, and even its median exceeds it.** This is the quantitative form of the argument made in the introduction: in this setting a one-run difference of one to three F1 points is not evidence about an architecture, and the practice of reporting one is what the protocol of this study is designed to replace.

D. Equivalence, and What Non-Significance Does Not Mean

Several comparisons in this paper end without a significant difference. A non-significant result is a statement about what the data can resolve, not a finding that two methods perform the same; concluding equivalence from a large p -value inverts the logic of the test. Where this paper needs to say that two operators are interchangeable, it tests that claim directly.

Two one-sided tests. For an equivalence bound δ , the two null hypotheses $H_{01} : \mu \leq -\delta$ and $H_{02} : \mu \geq +\delta$ are each tested one-sided, and $p_{\text{TOST}} = \max(p_1, p_2)$. Rejecting both places the difference inside $(-\delta, +\delta)$, which is equivalent to the $(1 - 2\alpha)$ confidence interval lying entirely within those bounds. Each comparison is reported as one of three outcomes: *different* when the difference is significant after Holm correction, *equivalent* when $p_{\text{TOST}} < \alpha$, and *undetermined* when neither holds. The third outcome is the honest description of most comparisons here, and we use that word rather than letting a large p -value imply the second.

The bound. δ was fixed at 0.02 F1 before the comparisons were run, and it was chosen from measurement noise rather than from the results. Two quantities set it: the run-to-run standard deviation under cross-validation, which ranges from 0.031 to 0.054 across configurations, and the mean replication error of an identical run reported above, 0.0195. A difference smaller than either of these is below what the apparatus can resolve, so treating it as a real architectural effect would be interpreting noise. Alongside each test we also report $\delta_{\min}$, the smallest bound at which equivalence could have been declared, so that a reader who prefers a different bound can apply it without re-running anything.

Overlapping training sets. In repeated cross-validation the training sets of different folds and repetitions overlap heavily—the same trip appears in the training portion of most partitions—so runs are not independent and the ordinary paired t -statistic underestimates the variance. The Nadeau–Bengio correction adds the overlap term,

$$\hat{\sigma}_{\text{corr}}^2 = \left(\frac{1}{n} + \frac{n_{\text{test}}}{n_{\text{train}}} \right) S^2, \quad (21)$$

where n is the number of runs and S^2 the sample variance of the paired differences. With the trip as the unit, $n_{\text{train}} = 59$ and $n_{\text{test}} = 18$, giving an overlap ratio of 0.305. On a pilot of two repetitions this widened the interval by a factor of 2.01 and

moved a comparison from $p = 0.174$ to $p = 0.482$ —which is the entire point of applying it. The correction is used for every cross-validated comparison, and for equivalence tests as well, where a wider interval makes an equivalence claim harder rather than easier.

What the fixed partition can and cannot show. On the fixed partition the same training set is used by every seed, so seed-level tests describe variation over initializations only and do not generalize to new data. Their resolution is also bounded combinatorially: with five seeds per configuration, an exact two-sample permutation test has $\binom{10}{5} = 252$ arrangements and a smallest attainable two-sided p -value of $2/252 = 0.0079$, which Holm correction over a family of four comparisons raises to 0.0317. A fixed-partition comparison can therefore reach significance only when the two sets of runs separate completely. This is a property of the design, not of any result obtained under it, and it is the reason the cross-validated analysis is treated as primary throughout.

1) *Pooling Uses One Seed, and What That Costs:* Under cross-validation each configuration is trained at two seeds per fold, but the *pooled* analyses—the subtype breakdown and the trip-level cluster tests—use the predictions of one seed only. Pooling both would enter every window twice and violate the independence the cluster resampling assumes at the trip level. The choice is deliberate, and it invites an obvious question: does the central negative result depend on which seed was pooled?

Repeating the entire pooled analysis with the second seed answers it in two parts. The decisions are unchanged: under either seed, none of the four hybrid variants differs significantly from the temporal-only control after Holm correction. The point estimates are not merely unchanged in magnitude but unstable in *sign*—in three of the four comparisons the difference against the control reverses direction between the two seeds, and pooled F1 for fixed concatenation moves from 0.8528 to 0.8885 on the seed alone.

We report this because it is a stronger statement than the one the paper would otherwise make. The claim is not only that these comparisons are not resolvable at this sample size; it is that their point estimates do not even retain their direction across two runs of the same configuration on the same partitions. Any reading of the pooled differences as small consistent effects would be reading run-to-run variation.

E. Data Provenance and What the Model Sees

A reviewer asked directly whether the graph, the node features or the topology carry information from the test trajectories, and whether the attention layer passes messages over a graph that includes test segments. Both questions have a yes in them, and the answer is more useful stated as a table than as a paragraph. Table 11 lists every input the model or the protocol touches, when it is fixed, and whether anything derived from the test partition reaches it.

The first three rows are the substantive admission. The road network is built once from the OpenStreetMap extract and held fixed, so every node—including those traversed only by test trips—is present in the graph during training, and the attention

Table 11. Provenance of every input. *Test access* asks whether the item is computed from, or influenced by, data in the test partition.

Item	Fixed when	Test access
Road topology (edges)	before any split	yes, by design
Node features	before any split	yes, by design
GAT message passing	every forward pass	yes, by design
Trajectory windows	before split, by trip	no
Train/val/test partition	once, seed 42, grouped by trip	—
Feature scalars	fitted on training trips only	no
Class weight w_{pos}	from training labels only	no
Checkpoint selection	validation F1 of the same run	no
Decision threshold	validation F1 of the same run	no
Graph construction choice	<i>not selected</i> ; both reported	no
Label-purity ceiling	training labels only	no

layer computes an embedding for all of them on every forward pass. The setting is **transductive**, and results should be read as such.

This is not by itself label leakage: no test *label* enters the graph, and the node features are static road attributes drawn from OpenStreetMap rather than from the trajectories. It does, however, mean that the model is never asked to embed a road it has not seen, and no result here supports a claim about generalisation to an unseen network. An inductive control, in which the training graph is restricted to training-trip nodes, is reported in Section VII-O.

The edge rule is simple enough to state in full. Each segment identifier has the form `<start node>–<end node>`, taken from the OpenStreetMap node pair it spans; a directed edge runs from segment *A* to segment *B* when *A*’s end node equals *B*’s start node. No edge is created from co-occurrence within a trip, from label similarity, or from any learned quantity. On the trip-instance graph this yields 53,660 edges over 4,678 nodes; merging duplicate traversals of the same physical road leaves 699 edges over 628 nodes.

The last two rows record the corrections described in Sections VI-A4 and V-B. Earlier versions of this work placed a test-derived quantity in both: a threshold averaged across folds, and a label-purity ceiling computed over all partitions and used to choose a graph. Neither remains.

F. Evaluation Metrics

Performance is assessed using accuracy, precision, recall, F1-score, ROC-AUC, average precision, and—for the first time in this line of work—probability calibration via the Brier score and Expected Calibration Error (ECE).

G. Implementation

Experiments were conducted with PyTorch 2.7.1 (CUDA 11.8) and PyTorch Geometric 2.7.0 on an NVIDIA RTX 4060 Laptop GPU (8 GB). All preprocessing, model, training, and evaluation code, together with the complete result files, is released (Section VIII-C).

VII. RESULTS AND DISCUSSION

A. Reference Points: Rule-Based and Classical Baselines

Because several anomaly subtypes are defined as threshold functions of the model’s own input channels (Table 8), we first

establish how much of the task is solvable without learning, and how much is solvable with classical feature-based learning. Table 12 and Fig. 2 report both alongside the hybrid model. All learned models use the fixed grouped partition, corrected normalization, validation-selected thresholds, and multi-seed reporting.

Table 12. **[Fixed partition, 5 seeds]** Reference points on the fixed test partition ($n = 4,799$, 157 positives)

Model	Precision	Recall	F1-Score	ROC-AUC
Rule: $v < 1$ only (0 params)	—	—	0.8129	—
Rule: $v < 1 \vee a > 3$ (0 params)	—	—	0.8317	—
Rule: 3 thresholds (0 params)	0.8354	0.8726	0.8536	—
XGBoost (window aggregates)	0.9200	0.8166	0.8651 ± 0.0052	0.9966
LightGBM (window aggregates)	0.9355	0.8127	0.8698 ± 0.0075	0.9966
Random Forest (window agrg.)	0.9518	0.8013	0.8699 ± 0.0061	0.9964
LSTM only (no spatial branch)	0.9400	0.7694	0.8455 ± 0.0190	0.9294
Hybrid LSTM-GAT	0.9336	0.8178	0.8713 ± 0.0188	0.9638

Three observations follow. First, a **zero-parameter classifier using three generation thresholds reaches $F1 = 0.8536$** , within 2 points of a 220,802-parameter neural model. Second, gradient-boosting models on 38 hand-crafted window aggregates reach 0.865–0.870 and attain **higher ROC-AUC than any neural configuration** (0.9966 versus 0.9638), indicating better threshold-independent ranking. Third, the hybrid model attains the highest mean F1, but by a margin of 0.0014 over Random Forest—roughly one tenth of its own seed-to-seed standard deviation.

None of these differences is statistically significant under paired testing (Table 13), including the comparison against the spatial-branch ablation. With 157 positive test instances distributed over roughly 13 trips (Section VI-B), the available power cannot resolve differences of one to three F1 points. We state the consequence directly: **on this partition the hybrid model is not distinguishable from a Random Forest on hand-crafted window aggregates, from a three-threshold rule with no parameters, or from itself with the spatial branch removed.** The cross-validated analysis of Section VII-E is where the remaining discriminative evidence lies.

Table 13. **[Fixed partition, window-level tests]** Paired comparisons against the hybrid LSTM-GAT model. The two $\Delta F1$ columns are *different quantities* and are reported separately: the first is the difference of five-seed means, the second is the difference on the median seed, which is the run the interval and the test are computed on. The interval is the quantity that brackets the median-seed difference, not the mean.

Comparison	$\Delta F1$ (5-seed mean)	$\Delta F1$ (median seed)	Bootstrap 95% CI (median seed)	McNemar p
vs. LightGBM	+0.0016	+0.0192	$[-0.0063, +0.0473]$	0.302
vs. XGBoost	+0.0063	+0.0280	$[+0.0034, +0.0558]$	0.049
vs. Random Forest	+0.0014	+0.0084	$[-0.0132, +0.0303]$	0.754
vs. 3-threshold rule	+0.0178	+0.0316	$[-0.0086, +0.0740]$	0.073
vs. LSTM only	+0.0258	+0.0081	$[-0.0147, +0.0316]$	0.549

Two corrections to the previous version of this table are worth recording, because both produced numbers that could not be reconciled with each other. First, the mean difference and the interval were placed in one row without stating that they are computed on different quantities—the mean over seeds and a single seed respectively—so a mean falling outside its neighbouring interval looked like an arithmetic error.

Second, and more consequentially, the paired bootstrap applied the *hybrid model’s* decision threshold to the competitor’s probabilities. Since the models are calibrated differently, this evaluated every competitor away from its own operating point and inflated the difference systematically in the hybrid’s favour: the Random Forest interval read $[+0.038, +0.116]$ against a mean difference of $+0.0137$. With each model scored at its own validation-selected threshold, the same interval is $[-0.0132, +0.0303]$, and the comparison is not significant. The values above use the corrected procedure. Both remain window-level statistics and are superseded by the trip-level analysis in Section VII-P.

Reference points: rule-based and classical models

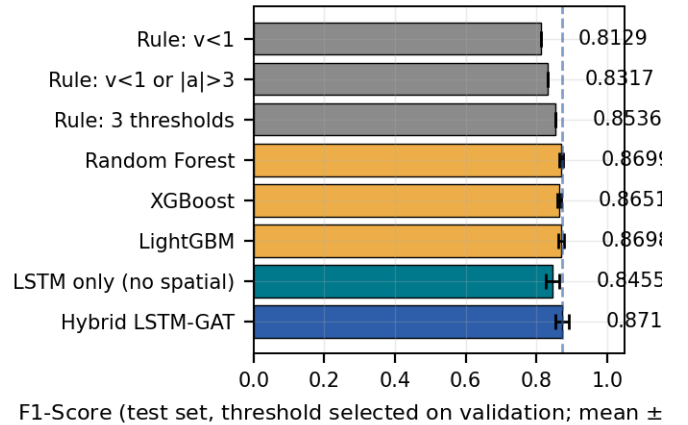


Figure 2. Reference points: zero-parameter rules and classical feature-based models against the hybrid neural model on the fixed test partition.

B. Encoder Benchmark

Reviewers of spatio-temporal anomaly detectors reasonably ask for comparison against STGCN [30], DCRNN [31], and Graph WaveNet [32]. These architectures target a structurally different formulation: every graph node carries a continuous time series and the model predicts jointly across all nodes at each step. The present task maps a single vehicle’s 50-step window to a single road segment for binary classification, so these models cannot be deployed unmodified.

We therefore isolate the **core operator** of each architecture and evaluate it under an identical task, budget, partition, fusion operator, and classification head. On the temporal axis we compare the stacked LSTM against a GRU (the DCRNN recurrent backbone), a gated temporal convolution (STGCN), a dilated causal convolution stack with residual and skip connections (Graph WaveNet), and a Transformer encoder. On the spatial axis we compare the GAT against Chebyshev graph convolution (STGCN), bidirectional diffusion convolution (DCRNN), a learnable adaptive adjacency (Graph WaveNet), and no spatial branch at all. This adaptation is a deliberate scoping decision and is stated as such.

Table 14 and Fig. 3 report the outcome. The fusion operator is held at static weighted summation throughout this benchmark. This choice follows from Section VII-C: the gated operator exhibits a collapse mode that would inject variance unrelated to encoder quality.

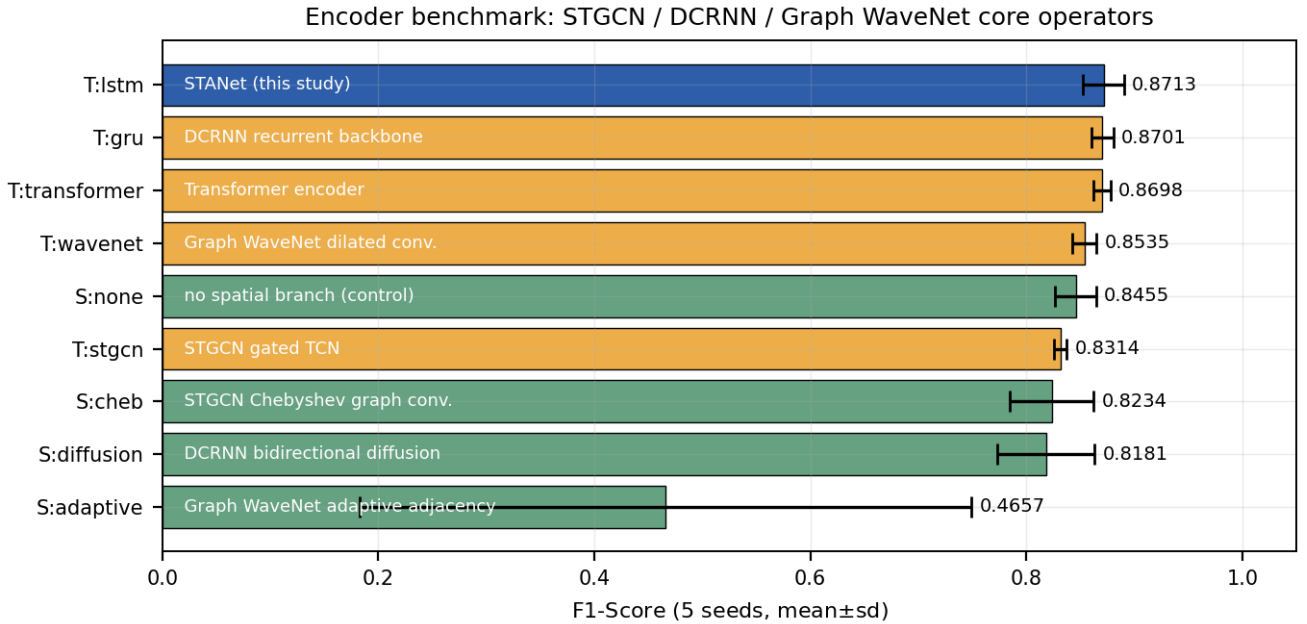


Figure 3. Encoder benchmark. Bars show mean F1 over five seeds; error bars show standard deviation.

Table 14. [Fixed partition, 5 seeds, window-level tests] Encoder benchmark, five seeds, identical protocol. The p -values are paired McNemar tests against the LSTM–GAT row; Table 35 repeats them at the trip level.

Config	Origin	F1-Score	ROC-AUC	Params	p
LSTM + GAT	this study	0.8713 ± 0.0188	0.9638	220,802	—
GRU	DCRNN backbone	0.8701 ± 0.0102	0.9929	170,626	0.118
Transformer	Transformer	0.8698 ± 0.0081	0.9899	292,098	0.815
Dilated causal conv	Graph WaveNet	0.8535 ± 0.0114	0.9799	416,258	0.115
No spatial branch	control	0.8455 ± 0.0190	0.9294	215,234	0.549
Gated TCN	STGCN	0.8314 ± 0.0058	0.9861	122,242	0.003
Chebyshev conv	STGCN	0.8234 ± 0.0389	0.9481	223,106	0.007
Diffusion conv	DCRNN	0.8181 ± 0.0450	0.9633	228,290	0.003
Adaptive adjacency	Graph WaveNet	0.4657 ± 0.2833	0.7891	321,850	<0.001

The LSTM–GAT pairing attains the highest mean F1 among the nine configurations, but the margin is significant against only four of the eight alternatives, and the identity of those four is the substantive result. The pairing separates cleanly from the STGCN gated temporal convolution ($p = 0.003$) and from all three alternative *spatial* operators—Chebyshev convolution ($p = 0.007$), diffusion convolution ($p = 0.003$), and adaptive adjacency ($p < 0.001$). It does *not* separate from any of the remaining strong temporal encoders: the GRU backbone (0.8701 ± 0.0102 , $p = 0.118$), the Transformer encoder (0.8698 ± 0.0081 , $p = 0.815$), or the dilated causal convolution stack (0.8535 ± 0.0114 , $p = 0.115$). The top three configurations lie within 0.0015 F1 of one another, an order of magnitude below the leader’s own seed standard deviation.

We draw the conclusion the measurement supports rather than the one the architecture was proposed to demonstrate: **the choice of temporal encoder is not resolvable on this data, while the choice of spatial operator is**. Attention-based aggregation is measurably better than the Chebyshev, diffusion, and adaptive alternatives; whether the recurrence is an LSTM, a GRU, or self-attention is not something these experiments can decide. The GRU reaches the same F1 with

23% fewer parameters and the highest ROC-AUC of any neural configuration, and would be the preferable choice on that basis alone.

Two negative results are informative. Replacing the GAT with Chebyshev or diffusion convolution falls *below* the no-spatial-branch control, indicating that graph message passing is not intrinsically helpful here and that any benefit is specific to attention-based aggregation. More strikingly, Graph WaveNet’s learnable adaptive adjacency collapses (0.4657 ± 0.2833): parameterizing a free $4,678 \times 4,678$ adjacency matrix is not identifiable at this data scale. Together with Section VII-C, a consistent pattern emerges—**freely parameterized adaptive components are unstable in this regime, whereas fixed or minimally parameterized alternatives are both stronger and more reliable**.

a) *Note on the previous version:* The margins reported here are smaller and fewer than in the earlier version of this work, which reported 0.8836 ± 0.0033 for LSTM–GAT and significance against the DCRNN backbone and both Graph WaveNet components. That table was computed over three seeds rather than five. Extending to five seeds raised the leader’s standard deviation from 0.0033 to 0.0188, and the comparisons against the GRU and dilated-convolution encoders ceased to be significant. A three-seed standard deviation of 0.0033 on a configuration whose five-seed spread is 0.0188 is not a small error bar; it is an under-sampled one.

C. Fusion Ablation

We now hold both encoders fixed and vary only the fusion operator, over five seeds. Table 15 and Fig. 4 summarize the outcome.

The sample-wise gate ranks **last of four**. Two aspects of this result matter more than the ranking itself.

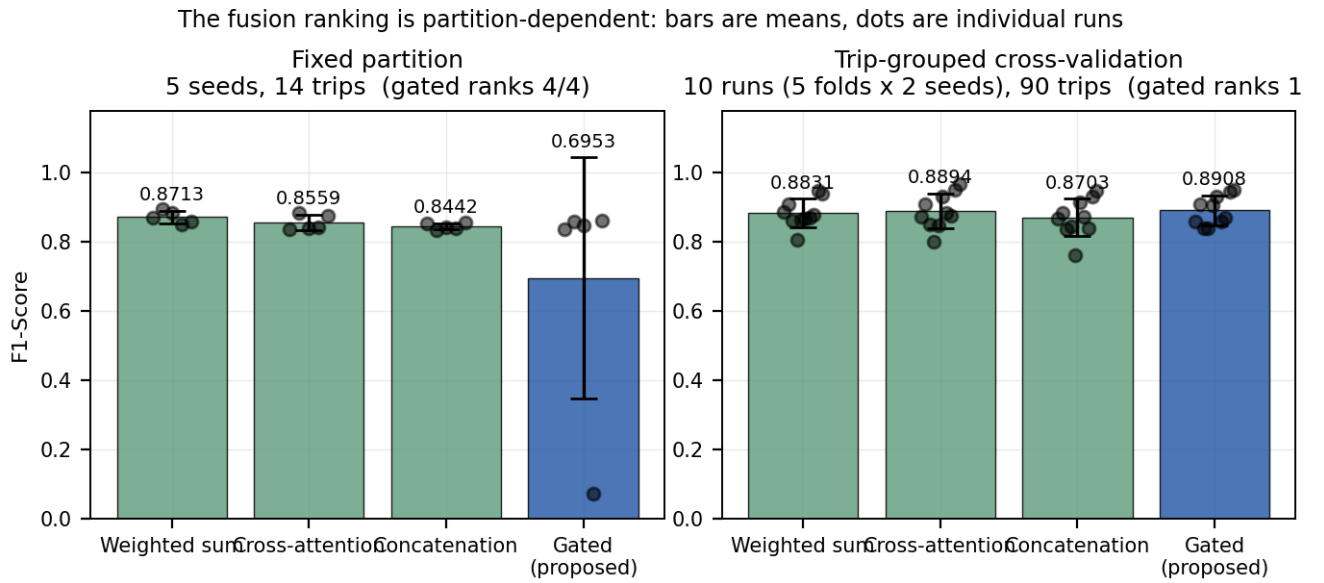


Figure 4. The fusion ranking is partition-dependent. Left: the fixed partition (five seeds, 14 trips), where the gated operator ranks fourth of four and one run collapses. Right: trip-grouped cross-validation (ten runs, 90 trips), where the same operator ranks first of four and no run collapses. Bars are means with standard deviation; dots are individual runs. Under cross-validation no pairwise difference is significant at the trip level after Holm correction (Section VII-E).

Table 15. **[Fixed partition, 5 seeds]** Fusion ablation, five seeds, all other components identical

Fusion	F1-Score	ROC-AUC	Worst seed	Params
Static weighted sum	0.8713 ± 0.0188	0.9638	0.8491	1
Cross-attention	0.8559 ± 0.0228	0.9417	0.8369	16,640
Fixed concatenation	0.8442 ± 0.0091	0.8993	0.8339	0
Gated (proposed)	0.6953 ± 0.3488	0.8752	0.0715	4,161

A degenerate collapse mode. In one of five seeds the gated model failed completely ($F1 = 0.0715$), with a mean g_{gate} of 0.026 and a mean of 0.0125 on anomalous samples. The gate had locked onto the spatial modality, which—as Section VII-B shows—cannot classify on its own, and training could not escape. Equation 16 admits this optimum by construction: as $g_{gate} \rightarrow 0$ the temporal signal is suppressed multiplicatively and its gradient vanishes with it. Concatenation has no such mode, because the classifier weights the two embeddings independently and can simply ignore the uninformative one. No collapse occurred in any of the fifteen runs of the other three operators.

This collapse is reproducible in the strongest sense available to us. The earlier version of this study observed it at seed 45 under a different cuDNN configuration ($F1 = 0.0833$). The experiments reported here were re-run from scratch under the protocol of Section VI-A4, and the collapse recurred *at the same seed* with a comparable magnitude. It is a property of the operator and this initialization, not an artifact of a particular backend setting.

The learned optimum is a constant. In the static weighted-sum variant, the single learnable scalar converged to $\{0.519, 0.522, 0.518, 0.498, 0.497\}$ across the five seeds—that is, to approximately equal weighting in every run. The data does not call for a sample-dependent mixing weight. The one-parameter static weight also scores higher on this partition,

though that difference is not resolvable here (Section VII-I); what is resolvable is the spread. Its seed standard deviation is ± 0.0188 against ± 0.3488 for the adaptive gate, a factor of eighteen, and a difference in stability of that size is visible where a difference in means is not.

Paired single-run tests are the wrong instrument for this comparison. On the median seed the test does not separate the gated model from concatenation ($p = 1.000$) or from cross-attention ($p = 0.727$), and its difference from the static weighted sum is -0.0379 with interval $[-0.0708, -0.0119]$; none of these single-partition numbers reflects the 0.35 standard deviation that is the substantive finding. Across-seed instability is not visible to a test that examines one seed. This is itself an argument for the multi-seed protocol of Section VI-A3.

Table 16. **[Fixed partition, 5 seeds]** Cost of the four fusion operators. Fusion parameters counts only the operator; total counts the whole model. Epochs and training time are means over the five fixed-partition seeds, on one NVIDIA RTX 4060 Laptop GPU. Latency is the median over a single window, measured on the trip-instance graph.

Fusion	Fusion par.	Total par.	Epochs	Train (min)	GPU (ms)	GPU cached	CPU (ms)
Static weighted sum	1	220,802	34.4	2.72	2.87	0.64	27.82
Fixed concatenation	0	222,849	19.8	1.50	2.56	0.55	23.14
Sample-wise gate	4,161	224,962	31.4	2.66	2.80	0.78	25.07
Cross-attention	16,640	237,441	30.0	2.49	2.91	0.96	19.83

Table 16 answers a question the ranking alone leaves open: what the four operators cost. The span in fusion parameters is four orders of magnitude, from none at all for concatenation to 16,640 for cross-attention, yet total model size varies by under 8% because the temporal branch dominates. Training time and single-window latency differ by less than the seed-to-seed variation in either. Cross-attention is the most expensive operator on every axis and finishes second of four under cross-validation and second on the fixed partition; the single scalar of the weighted sum is the cheapest learnable option and finishes third and first respectively. Neither ordering is

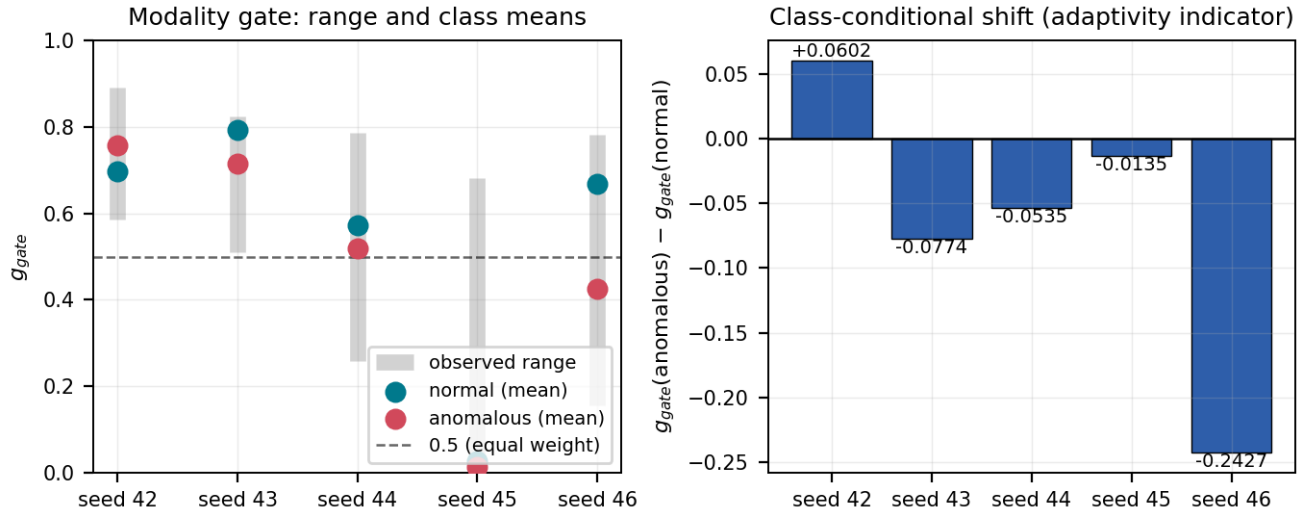


Figure 5. Modality gate behavior across five seeds. Left: observed range and class means. Right: class-conditional shift, which is inconsistent in both magnitude and sign.

significant (Section VII-E), which is the point: the operators differ in cost far more than the measurements can distinguish them in accuracy.

D. Gate Behavior

Section IV-D established that gating can only outperform concatenation through genuinely sample-dependent behavior. Fig. 5 examines whether such behavior is observed, by measuring the class-conditional shift $\bar{g}_{\text{anomalous}} - \bar{g}_{\text{normal}}$ across seeds.

The measured shifts are +0.0602, −0.0774, −0.0535, −0.0135, and −0.2427. Both the magnitude and the *sign* vary across seeds. A class-conditional interpretation of the gate—for instance, that anomalous samples shift weight toward the temporal branch—is therefore not a stable property of the trained model but an observation that holds in one run and reverses in the other four.

This is a correction to the interpretation offered in the earlier version of this work, which reported a shift from 0.70–0.75 on normal samples to approximately 0.80 on anomalous samples based on a single checkpoint. Re-evaluating the archived checkpoint gives class means of 0.860 and 0.856 respectively—a difference of 0.004, with the gate never falling below 0.5 on any sample. We report this explicitly because interpretability claims derived from one training run are exactly the kind of evidence this study argues against.

E. Cross-Validated Evaluation and Subtype Stratification

The fixed partition of Table 9 contains 157 positive test instances drawn from 14 trips. Section VII-P showed that this supports no inferential claim. We therefore repeat the entire evaluation—all four fusion operators and the spatial-branch control—under five-fold trip-grouped cross-validation crossed with five seeds (42–46), giving twenty-five runs per configuration. Reviewer 3 asked specifically that the seed at which the fixed partition collapsed be included here rather than

left out of the cross-validated design, and it is. Every sequence receives an out-of-fold test prediction, so all **90 trips** enter the analysis rather than 14, and fold-wise test prevalence ranges from 3.40% to 10.49%. This is the primary evidence base of the paper.

Table 17. [Cross-validation, 5 folds $\times$ 5 seeds] Five-fold trip-grouped cross-validation crossed with five seeds, 42–46 (twenty-five runs per configuration, all 28,090 sequences). Collapse counts runs with F1 < 0.30. The pooled, Brier, and ECE columns are computed from one seed and are therefore unchanged from the two-seed design; the run columns are not.

Model	Run F1	Worst	Collapse	Pooled F1	Brier	ECE
LSTM only (control)	0.8840 $\pm$ 0.0380	0.8302	0/25	0.8901	0.0208	0.0548
Weighted-sum hybrid	0.8816 $\pm$ 0.0422	0.8062	0/25	0.8891	0.0236	0.0409
Concatenation hybrid	0.8794 $\pm$ 0.0440	0.7621	0/25	0.8528	0.0254	0.0560
Gated hybrid	0.8608 $\pm$ 0.1464	0.1875	1/25	0.8849	0.0257	0.0579
Cross-attention hybrid	0.8234 $\pm$ 0.2111	0.0869	2/25	0.8838	0.0267	0.0627

The fusion ranking depends on how many seeds are averaged. The gated operator finished last of four on the fixed partition, *first* of four when cross-validation was run over two seeds, and fourth of four over five. The temporal-only control, which has no spatial branch at all, finishes first. Three orderings from the same data, the same folds, and the same protocol—the only difference being how many seeds enter the mean. A study that reports a fusion ranking without stating how many seeds it averaged has not reported enough to be checked.

Neither ordering should be reported as a result, and we report neither. At the trip level, gating differs from the weighted sum by −0.0042 (95% CI [−0.0162, +0.0102], Holm $p = 1.000$), from cross-attention by +0.0011 (95% CI [−0.0120, +0.0175], Holm $p = 1.000$), and from concatenation by +0.0321 (95% CI [+0.0052, +0.0689], Holm $p = 0.099$). **No fusion operator is distinguishable from any other** once the correct unit and multiplicity correction are applied. Over all twenty-five runs the four hybrids span 0.0582 F1, which exceeds the 0.0414 replication error of a single run and might be read as an architectural effect. It is not one. Three of the 125 runs are degenerate, and all three belong to the two operators that learn a mixing weight; removing them leaves a span of **0.0095**

F1—under a quarter of the replication error—and within-configuration standard deviations of 0.037–0.044 for all five configurations alike. **The apparent architectural spread is the footprint of three failures, not a difference in what the operators learn when they train successfully.** The two-seed design could not see this: it produced no degenerate run at all, and so reported a tight ranking that the wider seed set dissolves.

The comparison against concatenation is the closest to significance (uncorrected $p = 0.033$), and it is also the one most affected by the pooling correction of Section VI-A4: concatenation’s pooled F1 falls by 0.0244 when each run is scored at its own threshold, more than any other variant. We note this rather than resting on it.

This is the paper’s central empirical statement, and it cuts in both directions. It does not vindicate the sample-wise gate: nothing here shows the gate helping. It equally does not condemn it: the fixed-partition result that placed it last, including its collapse, is not reproduced when five partitions are used instead of one. **What the data supports is that the choice of fusion operator in this setting is not resolvable—**and that a study reporting a ranking from a single partition, as the earlier version of this work did, would have reported one of two opposite orderings depending on which partition it happened to use.

The collapse mode of Section VII-C remains a real and reproducible phenomenon, and with five seeds it appears in the primary cross-validated table rather than only on the fixed partition: three of the fifty runs belonging to weight-learning operators are degenerate, against zero of seventy-five for the fixed-weight operators and the control. An earlier version of this paper reported that *none of ten* runs collapsed, which was a statement about ten runs rather than about the phenomenon; those ten runs were two seeds, and neither was the seed at which the fixed partition had failed. The repeated-partition campaign of Section VII-F puts a rate on it instead, and the rate separates the operators cleanly (Table 21). It is not evidence that one operator is generally worse than another; it is evidence about a class of operators.

The spatial-branch control is equally instructive: removing the spatial branch entirely yields 0.8840 ± 0.0380 , the highest mean and the smallest standard deviation of the five, and no test reported here separates it from any hybrid variant. Following Section VI-D, that is an undetermined comparison rather than a demonstration that the branch contributes nothing. On the fixed partition the same ablation appeared to cost 3.5 F1 points at $p = 0.035$. **We withdraw that claim.** Section VII-E1 examines whether the spatial branch helps anywhere at all.

1) *Subtype Stratification:* Section VII-A identified speed-limit violation as the one anomaly subtype whose defining rule depends on a quantity ($v_{\max}$) available only to the spatial branch. Pooling out-of-fold predictions gives 144 such sequences drawn from all 90 trips—enough for a stratified comparison that the single partition, with 13 positives from 14 trips, could not support. Table 18 and Fig. 6 report the result, with all tests at the trip level and Holm correction across the three subtypes.

No comparison in this table is significant at the trip level

after Holm correction, and the table is worth reading carefully because an earlier version of this work reported one that was.

The predicted effect is absent. On speed-limit violations—the subtype whose generating rule is a function of $v_{\max}$, a quantity present only in the node features—the hybrid model is *worse* than the temporal-only model (0.1508 versus 0.2288), and the interval $[-0.2986, +0.1219]$ spans zero. Both are weak in absolute terms. We verified that the label rule and the model input use the same speed limit: applying the documented rule $v > 1.10 v_{\max}$ directly to the node feature recovers all 13 positive windows in the fixed test partition. Five of those thirteen lie on segments whose speed limit is an author-defined default, so the circularity noted in Section VII-S is real and, for this subtype, rests partly on an imputed value. The branch nonetheless fails to exploit the feature.

The unpredicted effect has also gone. Abnormal dwell was reported in the previous version as a gain of +0.0335 F1 with a Holm-corrected p of 0.002, the study’s only comparison significant at the trip level. It was an artifact of the pooling defect described in Section VI-A4: the pooled analysis applied the mean of all fold-seed thresholds to every prediction rather than each run’s own. With each run scored at its own validation threshold the gain is +0.0074, with an interval of $[-0.0084, +0.0268]$ that contains zero and a Holm-corrected p of 0.814. **We withdraw that finding.**

What remains is a hint in the opposite direction. Signal freeze has an uncorrected trip-level p of 0.0431 and an interval of $[-0.1238, -0.0065]$ that excludes zero—but the sign is negative: the hybrid is *worse* than the control (0.0261 versus 0.0848). After correction across the three subtypes it is not significant (0.129), and with 62 positive windows we do not claim it. We report it because it is now the comparison closest to significance in the study, and reporting only the directions that favour the proposed architecture is the practice this paper argues against.

Across all three subtypes, therefore, **no evidence remains that the spatial branch contributes to detection.**

F. Repeated Grouped Partitions

Cross-validation as reported above uses one grouped partition of the 90 trips. GroupKFold is deterministic, so that partition is a single draw, and every cross-validated number in this paper is conditional on it. To separate what depends on the partition from what depends on the architecture, the entire design was repeated over five partitions: the trip list is permuted with a partition-specific seed and split into five equal blocks, which preserves the grouping—no trip’s windows are ever split across folds—while changing which trips are tested together. Five partitions $\times$ five folds $\times$ two seeds $\times$ five configurations $\times$ two graph constructions gives 500 runs, all under one protocol stamp. The first partition reproduces the ordinary GroupKFold split and reuses its cached runs, so the first slice of this design is Table 17 rather than a second set of numbers for the same models.

Uncertainty. Training sets overlap heavily across folds and partitions, so paired differences are analysed with the Nadeau–Bengio correction of Eq. 21. With 18 test trips against 59

Table 18. [Cross-validation, pooled, trip-level tests] All anomaly subtypes present in the data, on pooled cross-validated predictions. Windows counts positive windows; Trips counts the trips containing at least one. Precision, recall and F1 are for the hybrid with weighted-sum fusion; the final columns compare it against the control without a spatial branch. Intervals and p -values are trip-level over the 90 trips and are reported only where the subtype has at least 5 positive trips and 30 positive windows; rows marked † are below that and are given descriptively. False positives are shared across rows: the negative set is held fixed and only the positive subset changes, so the hybrid records 89 false positives against every subtype and differences in F1 arise from recall. Subtype 6 of the taxonomy (GPS speed artefact) does not occur in the generated data.

Subtype	Windows	Trips	Precision	Recall	F1	LSTM-only F1	Δ	Trip 95% CI	p
Abnormal dwell	1,154	33	0.9250	0.9515	0.9381	0.9307	+0.0074	[−0.0080, +0.0265]	0.406
Crash	569	7	0.8578	0.9438	0.8987	0.8935	+0.0052	[−0.0519, +0.1054]	0.852
Speed-limit violation	144	28	0.1759	0.1319	0.1508	0.2288	−0.0780	[−0.3044, +0.1152]	0.506
Signal freeze	62	24	0.0220	0.0323	0.0261	0.0848	−0.0587	[−0.1283, −0.0046]	0.043
Abnormal heading change†	21	13	0.0220	0.0952	0.0357	0.0336	+0.0021	—	—
Abnormal acceleration†	19	15	0.0111	0.0526	0.0183	0.0342	−0.0158	—	—
GPS acceleration artefact†	8	8	0.0111	0.1250	0.0204	0.0377	−0.0173	—	—
GPS heading artefact†	8	6	0.0000	0.0000	0.0000	0.0000	+0.0000	—	—

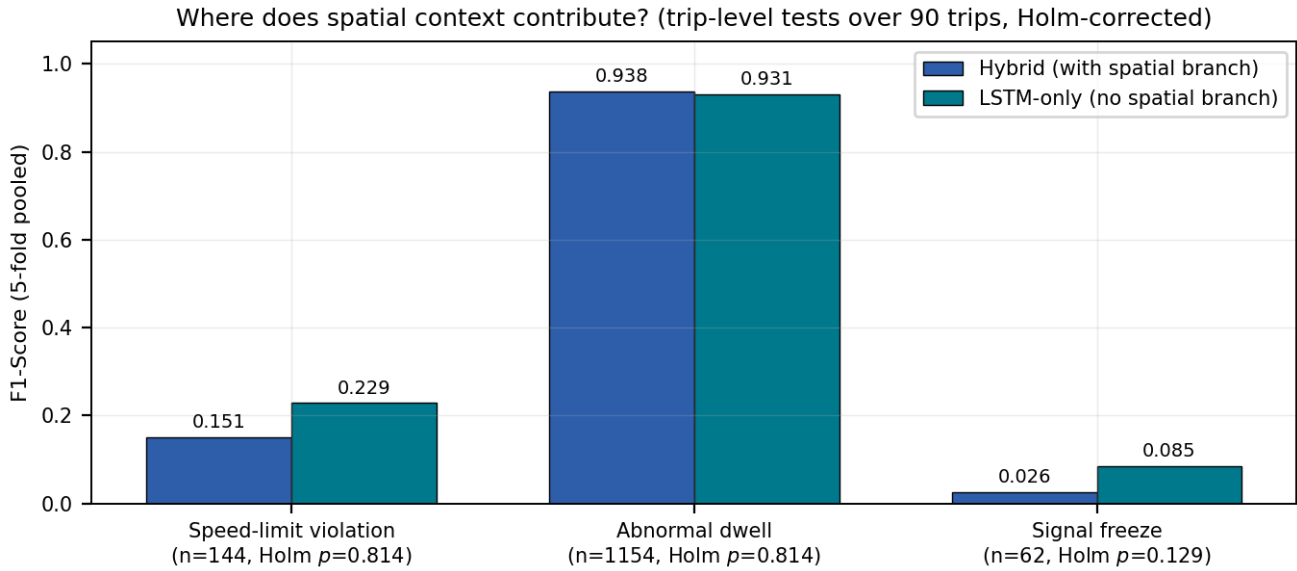


Figure 6. Subtype stratification on pooled cross-validated predictions. The spatial branch helps where it was not predicted to (abnormal dwell) and does not help where it holds the decisive feature (speed-limit violation).

Table 19. [Repeated cross-validation, 5 partitions $\times$ 5 folds $\times$ 2 seeds = 50 runs per cell] Every configuration under both graph constructions. The worst run is reported because the mean alone hides the collapse mode of Table 21.

Fusion operator	Trip-instance graph		OSM-merged graph	
	F1 (mean $\pm$ s.d.)	Worst	F1 (mean $\pm$ s.d.)	Worst
Gated (proposed)	0.8612 $\pm$ 0.1261	0.0795	0.8342 $\pm$ 0.1708	0.1063
Weighted sum	0.8845 $\pm$ 0.0443	0.7519	0.8812 $\pm$ 0.0474	0.7149
Concatenation	0.8715 $\pm$ 0.0670	0.5154	0.8811 $\pm$ 0.0400	0.7950
Cross-attention	0.8304 $\pm$ 0.1986	0.0789	0.8510 $\pm$ 0.1535	0.1039
No spatial branch	0.8796 $\pm$ 0.0386	0.7902	0.8796 $\pm$ 0.0386	0.7902

training trips the correction widens intervals by a factor of 4.03, and it changes conclusions rather than decorating them: on the merged graph three comparisons sit at uncorrected p of 0.0526, 0.0545 and 0.0566—all of which a reader would take as marginal—and move to 0.62 and above once the shared training data is accounted for.

Result. None of the twenty paired comparisons—ten within each graph construction—is significant after Holm correction, and none is equivalent at the pre-specified bound of 0.02 F1

either. Both statements matter. The tightest comparison, fixed concatenation against the temporal-only control on the merged graph, would support an equivalence claim at a bound of 0.0262; the honest summary is therefore that these operators differ by less than about 0.026 F1, not that they are the same. The spatial branch’s contribution is the most precisely estimated quantity here and the smallest: +0.0049 F1 on the trip-instance graph and +0.0016 on the merged graph, with intervals spanning zero in both.

The graph construction is not a hidden variable. Treated as the prespecified factor it is, rather than as a choice, it moves nothing: paired within each cell, the difference between constructions is between -0.0269 and $+0.0205$ F1 for the four hybrids and exactly zero for the temporal-only control, which has no spatial branch to be affected and therefore serves as the design’s own consistency check.

Table 20. [Repeated cross-validation, 50 runs per row] Where the variation in F1 comes from, as a percentage of total variance. The seed is the smallest identified source in every row.

Graph	Operator	s.d.	Part.	Fold	Seed	Resid.
Trip-instance	Gated (proposed)	0.1261	14.6	7.9	0.4	77.1
Trip-instance	Weighted sum	0.0443	10.5	1.6	1.1	86.8
Trip-instance	Concatenation	0.0670	8.6	8.8	1.9	80.7
Trip-instance	Cross-attention	0.1986	2.6	18.9	0.0	78.5
Trip-instance	No spatial branch	0.0386	6.9	4.9	0.0	88.2
OSM-merged	Gated (proposed)	0.1708	13.8	5.8	4.9	75.5
OSM-merged	Weighted sum	0.0474	6.9	8.8	0.1	84.2
OSM-merged	Concatenation	0.0400	2.7	2.6	0.3	94.4
OSM-merged	Cross-attention	0.1535	12.9	11.6	0.5	75.0
OSM-merged	No spatial branch	0.0386	6.9	4.9	0.0	88.2

Where the variation lives. Decomposing F1 over the three design factors (Table 20) gives the partition 8.6% of the variance on average, the fold 7.6%, and the seed 0.9%; the remainder is interaction and run-to-run noise, which a design with one observation per cell cannot separate further. The ordering is the point. The earlier version of this study varied the seed and held the partition fixed, which is to say it sampled the smallest identified source of variation while holding constant a larger one. This is the quantitative case for repeated partitions, and it is why the single-partition results of Section VII-A are reported here as a secondary view.

1) The Collapse Mode Belongs to a Class of Operators:

Eight of the 500 runs are degenerate ($F1 < 0.30$), and they are not distributed the way the earlier account implied (Table 21). Every one occurs in an operator that learns a mixing weight—the sample-wise gate or multi-head cross-attention—at a rate of 4.0%, against 0.0% across 300 runs of fixed concatenation, static weighted summation, and the temporal-only control. Nor is it confined to one graph: the gate collapses in 3 of 50 runs on the merged graph against 1 of 50 on the trip-instance graph. A previous version reported the opposite, that the collapse disappeared on the merged graph, on the basis of five seeds on one partition; with fifty runs per cell that reading does not hold.

Table 21. [Repeated cross-validation, 500 runs] Degenerate runs ($F1 < 0.30$). Every collapse belongs to an operator that learns its mixing weight.

Fusion operator	Learns w.	Trip-inst.	OSM-mrg.	Total
Gated (proposed)	yes	1/50	3/50	4/100
Weighted sum	no	0/50	0/50	0/100
Concatenation	no	0/50	0/50	0/100
Cross-attention	yes	3/50	1/50	4/100
No spatial branch	no	0/50	0/50	0/100

Mechanism. The gate was instrumented to record its distribution over the validation set after every epoch, under eval

mode and without gradients, so the measurement consumes no randomness and the protocol is unchanged. Ten runs were instrumented: five seeds on each graph construction.

What separates a collapsed run from a healthy one is whether the gate escapes towards the temporal branch. The fusion output is $g h'_{temp} + (1-g) h'_{spat}$, so $\bar{g} > 0.5$ means the representation is temporal-dominant. In nine of the ten runs the gate crosses that line, between epochs 4 and 19, and validation F1 rises at about the same time; all nine finish between 0.7557 and 0.8742. The tenth never crosses: its mean peaks at 0.3815, and it stops at $F1 = 0.0759$ when early stopping ends it at epoch 18. The gap is clean—the lowest peak among the nine is 0.7402, against 0.3815 for the one that fails—and Figure 7 shows it.

An earlier reading of this experiment attributed the collapse to the sign of the gate’s class-conditional separation, which is negative from the second epoch in the failing run. Examining all ten runs does not support that. On the merged graph two healthy runs have separations of -0.1900 and -0.3380 at epoch two, far more negative than the -0.0009 of the collapsed run, and both reach $F1 \approx 0.86$. The sign is not what decides the outcome; the escape is.

This supports the mechanism the earlier version proposed—the gate remains attached to an uninformative spatial branch—and replaces an inference drawn from two runs with a separation measured across ten. The failure is not a property of the proposed operator. It is what can happen when a fusion stage is free to learn how much of an uninformative branch to admit, and it is worth reporting for that reason.

G. What the Spatial Branch Contributes

Removing the spatial branch entirely, as in Section VII-E, cannot say *why* it makes no difference. The branch supplies two distinct things: a set of node features describing each road segment, and message passing over the road topology that mixes a segment’s representation with its neighbours’. An ablation that deletes both at once leaves the attribution open.

Three controls separate them. Each is trained from scratch under its own condition rather than applied to a trained model at inference, because the question is what the branch *contributes*, not how robust it is to perturbation. Removing the edges and keeping only self-loops leaves the node features intact but reduces the graph layer to a per-node transformation, so no message passing occurs. Shuffling the node feature rows across segments preserves the topology and the degree distribution while destroying the correspondence between a segment and its description. Replacing the edges with random pairs of the same density preserves both the features and the amount of message passing while discarding the road network itself.

None of the three differs from the reference after correction, and the point estimates run the wrong way for two of them: removing message passing scores $+0.0094$ F1, and shuffling the node features $+0.0123$. Replacing the real topology with random edges of matched density costs -0.0006 . Read together, neither the features nor the message passing carries measurable signal here, and the road network is interchangeable with an arbitrary graph of the same density.

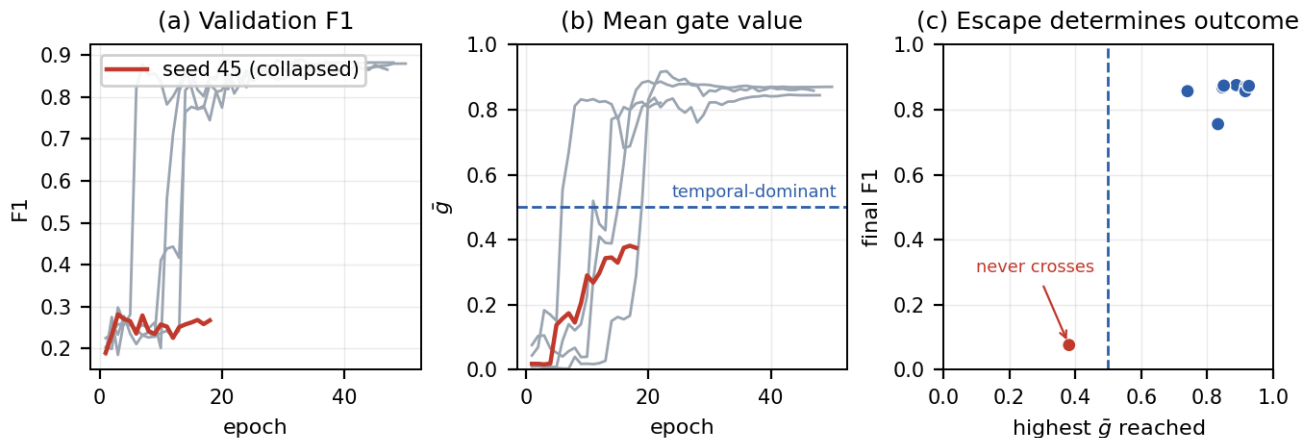


Figure 7. **[Fixed partition, 5 seeds per graph construction, measured every epoch]** Why one run collapses. (a) and (b) show the five seeds of the trip-instance graph, with the collapsed run in red. The gate must cross $\bar{g} = 0.5$ for the representation to become temporal-dominant; the collapsed run never does. (c) pools all ten instrumented runs: the highest mean gate value a run reaches separates the outcomes completely.

Table 22. **[Cross-validation, 5 folds $\times$ 2 seeds = 10 runs per row]** What the spatial branch contributes. Each condition is trained from scratch under that condition; the comparison against the reference is paired within run and corrected for overlapping training sets.

Condition	Edges	F1 (mean $\pm$ s.d.)	Δ vs reference	95% CI	Holm p
Reference (real features, real topology)	53,660	0.8831 ± 0.0412	—	—	—
Edges removed (self-loops only)	4,678	0.8925 ± 0.0370	+0.0094	$[-0.0068, +0.0256]$	0.664
Node features shuffled	53,660	0.8954 ± 0.0323	+0.0123	$[-0.0205, +0.0451]$	0.838
Random edges, same density	53,660	0.8825 ± 0.0348	-0.0006	$[-0.0279, +0.0267]$	0.962

The pooled trip-level tests reach significance for two of the three conditions—shuffled features at Holm $p = 0.0094$ and random edges at 0.0495—but these use a single seed’s predictions, and Section VI-D1 shows that pooled differences on this data do not retain their sign across seeds. The run-level comparison above uses both seeds and accounts for the overlap between training sets, and we base the conclusion on it. Both are in the released result files.

This is the most direct evidence in the paper for the scope stated in Section VII-R: on this dataset the task is solved from the kinematic channels, and the graph is not contributing to the solution. Whether that is a property of hybrid architectures or of a generator whose anomalies are defined almost entirely by kinematic thresholds is a question this dataset alone cannot settle; Section VII-L addresses the first half of it on a second road network.

H. Classical Baselines on Equal Footing

In the previous version the gradient-boosting baselines and the threshold rule were evaluated only on the fixed partition while the neural models were additionally cross-validated, so the numbers placed side by side came from different evaluation designs. All of them now run inside the same folds, with each fold’s own validation-selected threshold applied to that fold’s test predictions, and with the trip as the unit of inference.

The ordering is not the one the architecture section would predict. Gradient boosting on window aggregates reaches 0.9195 ± 0.0307 against 0.8608 ± 0.1464 for the proposed gated hybrid, and the zero-parameter threshold rule reaches 0.8938 ± 0.0375 —**a higher mean than every learned**

model in the table, including the temporal-only control at 0.8840 ± 0.0380 . The rule’s margin over the learned models is not significant at the trip level (+0.0171, Holm $p = 0.2288$), so the claim here is not that thresholds win; it is that four hand-written thresholds and no training are inside the interval of everything that was trained. Whatever the hybrid architecture is doing on this dataset, four window statistics and a gradient-boosted tree ensemble do at least as well.

The two levels of analysis disagree, and the disagreement is the point. Pooled at the trip level, LightGBM exceeds the proposed model by +0.0409 F1 with a cluster interval of $[+0.0278, +0.0552]$ and Holm $p = 0.0002$; XGBoost and the random forest are significant too. At the run level, pairing the same ten fold-seed runs and correcting for overlapping training sets, the same comparison is +0.0272 with interval $[-0.0106, +0.0650]$ and Holm $p = 1.000$. Neither number is wrong. The trip-level test conditions on one seed’s predictions and resamples trips, so it describes the sampling of trips only; the run-level test additionally carries seed and partition variability but has ten observations. We report both and draw the conclusion from the run-level test, consistently with Section VI-D1: the direction is reproducible, the magnitude is not resolvable, and a reader should not take +0.0409 as an estimate of how much better gradient boosting is.

What the comparison does support is narrower and still useful. There is no evidence here that the additional capacity of the hybrid architecture buys accuracy over a standard tabular baseline on aggregated window features, and the burden of showing otherwise falls on a design that can resolve differences of this size.

Table 23. [Cross-validation, 5 folds $\times$ 5 seeds; identical folds and threshold policy for every row] Learned and classical models on equal footing. The last two columns compare each classical model with the proposed gated hybrid at the trip level, pooled over all 90 trips.

Model	F1 (mean $\pm$ s.d.)	Pooled F1	Δ vs proposed	Holm p
Gated hybrid (proposed)	0.8608 $\pm$ 0.1464	0.8849	—	—
Weighted-sum hybrid	0.8816 $\pm$ 0.0422	0.8891	—	—
Cross-attention hybrid	0.8234 $\pm$ 0.2111	0.8838	—	—
Concatenation hybrid	0.8794 $\pm$ 0.0440	0.8528	—	—
Temporal only (no graph)	0.8840 $\pm$ 0.0380	0.8901	—	—
LightGBM on window aggregates	0.9195 $\pm$ 0.0307	0.9258	+0.0409	0.0002
XGBoost on window aggregates	0.9095 $\pm$ 0.0327	0.9156	+0.0307	0.0002
Random forest on window aggregates	0.9082 $\pm$ 0.0299	0.9106	+0.0257	0.0082
Three-threshold rule (zero parameters)	0.8938 $\pm$ 0.0375	0.9020	+0.0171	0.2288

Table 24. [Equivalence bound $\delta = 0.02$ F1, fixed before the comparisons] Outcome of every comparison the paper describes as showing no difference. *Different* means significant after Holm correction; *equivalent* means the difference lies inside $\pm\delta$; *undetermined* means neither. The last column is the smallest bound at which the tightest comparison in that family could have been declared equivalent.

Family	Structure	n	Different	Equivalent	Undetermined	Smallest δ
Repeated CV, trip-instance graph	all pairs	10	0	0	10	0.0352
Repeated CV, OSM-merged graph	all pairs	10	0	0	10	0.0262
Cross-validation, proposed vs others	reference	4	0	0	4	0.0316
Fixed split, fusion operators	reference	3	0	0	3	0.4787
Fixed split, temporal encoders	all pairs	10	4	1	5	0.0133
Fixed split, spatial operators	reference	4	1	0	3	0.0450

I. Which Comparisons Are Resolved, and Which Are Not

Section VI-D defines three outcomes. Table 24 reports which one each comparison in this paper receives.

Of the forty-one comparisons, five are resolved and one is equivalent; the remaining thirty-five are undetermined. Every comparison under repeated cross-validation—the paper’s primary design, with fifty runs per configuration on each of two graph constructions—falls in the last category. This is the result the paper rests on, and it is worth stating in the form the test actually supports: the architectural differences on this dataset are neither demonstrated nor shown to be absent.

The single equivalence is between a GRU and a Transformer temporal encoder, which differ by +0.0003 F1 over five seeds with $p_{\text{TOST}} = 0.0161$. It is the only comparison anywhere in the study tight enough to place inside a bound of 0.02 F1. The comparison of either against the LSTM is not: at +0.0012 and +0.0015 F1 the point estimates are just as small, but the intervals are wider and the correct statement is that the design cannot separate them.

The smallest bound column is the practical one. Under repeated cross-validation on the merged graph, the tightest comparison—fixed concatenation against the temporal-only control—would support an equivalence claim at $\delta = 0.0262$. On the trip-instance graph the corresponding figure is 0.0352. A reader who considers differences below three F1 points practically irrelevant can therefore read these results as establishing equivalence; a reader who does not, cannot. We report the number rather than choosing for them.

The fixed-partition fusion family is a separate case. Its smallest attainable bound is 0.4787, two orders of magnitude larger than anywhere else, because three of its four operators include runs that collapse and a mean difference over a bimodal

distribution carries almost no information. The collapse is the finding there, not the mean, and Section VII-F1 treats it as such.

1) *Transductive and Inductive Settings:* Table 11 records that the graph is supplied in full during training, so message passing runs over nodes belonging to held-out trips. No test label is ever used, but the topology and the node features of test segments do contribute to the representation. This is a transductive setting, and a detector moved to an unseen road would not have it.

The cost is measurable, so we measured it. For each fold, the nodes that belong *only* to test trips are isolated and their feature rows zeroed before training; a segment that also appears in a training or validation trip is left alone, because it is genuinely part of the training data. At inference the full graph is restored, which is what makes the setting inductive rather than merely degraded: the model must generalise to segments it never saw. Isolating rather than deleting keeps the node indices stable, so both settings train the same architecture with the same parameter count.

Neither configuration shows a measurable cost, and the two point estimates have opposite signs: the weighted-sum hybrid scores +0.0182 F1 in the inductive setting and the gated hybrid -0.0186 . Both intervals span zero comfortably. Given Section VII-G, this is what one would expect—if neither the node features nor the message passing carries measurable signal, then hiding some of the nodes cannot cost anything measurable either.

The result closes the leakage question in one direction only. It shows that the transductive setting is not inflating the reported numbers on this dataset. It does not show that a model trained here would transfer to a different road network, which is a

Table 25. [Cross-validation, 5 folds $\times$ 2 seeds = 10 runs per cell] Cost of hiding the test segments during training. Between 595 and 657 nodes per fold are isolated and their features zeroed while training; the full graph is restored for inference.

Configuration	Transductive	Inductive	Δ	95% CI	Holm p
Weighted-sum hybrid	0.8831 ± 0.0412	0.9013 ± 0.0315	+0.0182	$[-0.0328, +0.0691]$	0.882
Gated hybrid (proposed)	0.8908 ± 0.0429	0.8723 ± 0.0584	-0.0186	$[-0.0728, +0.0356]$	0.882

Table 26. [Cross-validation, 5 folds $\times$ 2 seeds, closed form] F1 at prevalences other than the observed one. Fold test partitions range from 3.40% to 10.49% anomalous, averaging 7.04%.

Configuration	$\pi = 1\%$	3%	8%	15%	25%	ROC-AUC
Gated hybrid	0.759	0.857	0.895	0.907	0.912	0.9676
Weighted-sum hybrid	0.730	0.837	0.885	0.900	0.907	0.9759
Temporal only	0.775	0.858	0.889	0.898	0.903	0.9423

Table 27. [Inference-time perturbation, threshold held fixed] F1 of the weighted-sum hybrid under speed noise of increasing magnitude (σ in km/h).

Noise model	$\sigma = 0$	0.5	1	2	5	10
White noise on speed	0.904	0.902	0.899	0.783	0.145	0.034
AR(1) noise on speed, $\rho = 0.9$	0.904	0.902	0.900	0.890	0.767	0.448

separate question and a separate experiment.

J. Sensitivity to Prevalence, Noise, and Imputation

Three properties of the dataset could plausibly carry the results: how often anomalies occur, how much sensor noise the inputs carry, and how the missing road attributes were filled in. Each is varied here.

1) *Anomaly prevalence*: Rather than subsample windows, which would add resampling noise and break the trip structure, prevalence is varied in closed form. True and false positive rates are computed within class and are therefore invariant to prevalence, so precision and F1 at any rate π follow exactly: $\text{precision} = \pi \text{TPR} / (\pi \text{TPR} + (1 - \pi) \text{FPR})$ with recall unchanged. These are not estimates; they are the same model at the same threshold under a different class balance. The threshold is held at the value each run selected on its own validation data, which is the realistic case: a deployed detector is not recalibrated for a prevalence it does not know.

F1 falls substantially as anomalies become rarer—by roughly 0.15 between 8% and 1%—which is expected, since precision depends on the class balance while ROC-AUC, also tabulated, does not change at all. The point for this paper is the second one: *the ordering of the configurations does not change at any prevalence*. The conclusion that no fusion operator is distinguishable is not an artefact of the prevalence this dataset happens to have.

2) *Sensor noise*: Noise is applied at inference to trained models. Because acceleration in this dataset is the first difference of speed (Section V-E), perturbing speed propagates into acceleration; adding independent noise to both would present a combination the generator cannot produce.

The two rows differ by an order of magnitude at the same σ , and the difference is a property of the differencing rather than of the model. White noise at one-second sampling produces acceleration noise of roughly 2 m/s² after differencing, which is the size of the anomaly thresholds themselves, so the acceleration channel is destroyed and F1 falls to 0.145 at

$\sigma = 5$. Real GPS speed error is correlated in time; under an AR(1) model with $\rho = 0.9$ the same σ costs 0.137 F1 and 2 km/h costs nothing measurable. Reporting only the white-noise curve would understate robustness; reporting only the correlated one would overstate it.

Bearing noise has no effect at all: F1 moves from 0.904 at 0° to 0.905 at 10°. The anomaly definitions are kinematic, and the detector runs on speed and acceleration; the heading channel carries no detection signal here, and the paper should not imply otherwise.

3) *Imputation of road attributes*: Roughly half of the node attributes are filled rather than measured: the lane count comes from a default for 46.2% of segments and the speed limit for 49.9%. A model could in principle be learning the fill pattern rather than the road.

None of the three alternatives differs measurably. Adding explicit indicators for what was imputed does not help, so the fill pattern is not carrying signal the model was exploiting. Replacing the defaults with the median over OSM-sourced segments of the same road type does not help either, so the conclusion does not depend on the fill rule. And removing the speed limit and lane count entirely costs 0.0112 F1 with an interval spanning zero, which bounds how much those two attributes contribute at all. This is consistent with Section VII-G: attributes that carry no measurable signal cannot be sensitive to how they were filled.

K. Window Length, Stride, and Baseline Tuning

1) *Window length and stride*: Every result above uses $\tau = 50$ with stride 10. The choice affects two things at once: how much history the model sees, and how dependent the observations are. Eight combinations were run, each written to its own protocol stamp so that cells cannot be confused with one another. The trip count is 90 in every cell, so trip-level comparisons remain valid across them; the window count is not, and is reported.

Table 28. [Cross-validation, 5 folds $\times$ 2 seeds = 10 runs per row] Four treatments of the imputed road attributes, each trained from scratch. Differences are paired against the released data and corrected for overlapping training sets.

Treatment	Node features	F1 (mean $\pm$ s.d.)	Δ vs released	95% CI
As released	17	0.8831 $\pm$ 0.0412	—	—
Plus imputation indicators	19	0.8764 $\pm$ 0.0480	−0.0067	[−0.0533, +0.0399]
Road-type median instead of default	17	0.8792 $\pm$ 0.0547	−0.0039	[−0.0764, +0.0686]
Speed limit and lanes removed	15	0.8719 $\pm$ 0.0502	−0.0112	[−0.0685, +0.0460]

Table 29. [Cross-validation, 5 folds $\times$ 1 seed per cell] Window length and stride. $\dagger$ marks the setting used throughout the rest of the paper. The last two columns are the spatial branch’s contribution, paired within run and corrected for overlapping training sets.

τ	Stride	Overlap	Test windows	LSTM + GAT	Temporal only	Δ spatial	p
25	10	60%	28,321	0.8990 $\pm$ 0.0384	0.8721 $\pm$ 0.0674	+0.0269	0.537
25	25	0%	11,350	0.7826 $\pm$ 0.2188	0.8694 $\pm$ 0.0460	−0.0868	0.579
50	10 $\dagger$	80%	28,090	0.8865 $\pm$ 0.0360	0.8871 $\pm$ 0.0468	−0.0006	0.981
50	25	50%	11,260	0.8519 $\pm$ 0.0485	0.8591 $\pm$ 0.0529	−0.0072	0.647
50	50	0%	5,654	0.7957 $\pm$ 0.0694	0.8441 $\pm$ 0.0569	−0.0483	0.354
100	10	90%	27,640	0.8706 $\pm$ 0.0536	0.9066 $\pm$ 0.0400	−0.0360	0.321
100	25	75%	11,080	0.8424 $\pm$ 0.0599	0.8625 $\pm$ 0.0553	−0.0201	0.694
100	50	50%	5,564	0.6246 $\pm$ 0.3594	0.6984 $\pm$ 0.2448	−0.0739	0.846

The spatial branch’s contribution is negative in seven of the eight cells and significant in none. The conclusion of Section VII-G therefore does not depend on the windowing: it holds at overlaps from 0% to 90% and at history lengths from 25 to 100 steps.

Two patterns are worth recording. Accuracy is not monotone in overlap—the best temporal-only cell is $\tau = 100$ with stride 10 at 0.9066 ± 0.0400 , and the worst is $\tau = 100$ with stride 50 at 0.6984 ± 0.2448 —and the variance tracks the number of windows rather than the overlap. Cells with roughly 5,600 test windows have standard deviations three to seven times larger than cells with 28,000, which is what one expects when the estimate is built from fewer observations. This is also where the hybrid becomes unstable: its two worst cells, 0.7826 and 0.6246, both carry standard deviations above 0.2.

2) *Were the baselines under-tuned?*: A comparison in which only the proposed model is tuned proves little. The learning rate was therefore searched over $\{5 \times 10^{-4}, 10^{-3}, 2 \times 10^{-3}\}$ for every configuration including the reference, with selection on validation F1 alone—the test partition plays no part in the choice—and the selected setting then confirmed over five seeds. The learning rate is inside the protocol stamp, so each search point has its own run store and the points cannot be mixed.

Table 30. [Fixed partition, 5 seeds at the selected rate] Symmetric learning-rate search. “Moved” indicates that the search selected something other than the default 10^{-3} .

Configuration	Rate	Moved	F1	Epoch
LSTM + GAT (ref.)	10^{-3}	no	0.8713 $\pm$ 0.0188	19.4
GRU temporal	10^{-3}	no	0.8701 $\pm$ 0.0102	25.6
Transformer temporal	10^{-3}	no	0.8698 $\pm$ 0.0081	23.2
Graph WaveNet temporal	10^{-3}	no	0.8535 $\pm$ 0.0114	21.0
STGCN temporal	10^{-3}	no	0.8314 $\pm$ 0.0058	14.0
Chebyshev spatial	10^{-3}	no	0.8234 $\pm$ 0.0389	24.0
Diffusion spatial	10^{-3}	no	0.8181 $\pm$ 0.0450	30.4
Adaptive adjacency	5×10^{-4}	yes	0.8113 $\pm$ 0.0719	8.2

The search changes the picture very little, which is the useful outcome. Seven of the eight configurations keep the default rate; only adaptive adjacency moves, and it remains the weakest. No configuration exhausted the epoch budget, so none was stopped before converging. After tuning, the GRU and the Transformer finish within 0.0015 F1 of the reference, and the three spatial alternatives remain between 0.0179 and 0.0600 below it.

The comparison in Section VII-I is therefore not an artefact of leaving competing operators at unfavourable settings. It also does not establish that these operators are weaker in general: each was adapted to this task and this input representation, and a different adaptation might behave differently.

L. An Independent Road Network, and the Price of a Leaked Feature

Every result to this point comes from one generator over one road network. A second generation campaign supplied datasets over a different network—110 trips across 1,183 distinct physical roads in Hamburg, against 90 trips across 628 in Kocaeli, with an average node degree of 9.1 against 11.5—together with a regenerated Kocaeli. Both arrived with a prescribed preprocessing filter, and auditing them before use turned out to matter.

1) *A label is determined by a graph feature*: In all thirty-six delivered datasets, every record on a tunnel segment carries the signal-freeze label: $P(\text{anomaly } 9 \mid \text{tunnel}) = 1.0000$ after the prescribed filter, which removes the exceptions. The tunnel flag is a node attribute, so it enters the spatial branch directly. The share of anomalies this covers differs sharply between the two networks: 60.4% of all anomalies in Kocaeli against 5.1% in Hamburg.

A model with graph access can therefore read a label off one binary input. Rather than discard the data or ignore the problem, we ran every configuration twice—with the tunnel attribute

Table 31. [Cross-validation, 5 folds $\times$ 2 seeds = 10 runs per cell] Both networks, with and without the leaking node attribute. The temporal-only control has no spatial branch and is identical in both settings by construction, which is the design’s internal check.

Configuration	Hamburg		Kocaeli	
	Leak removed	Leak present	Leak removed	Leak present
Gated hybrid (proposed)	0.8664 ± 0.0383	0.6435 ± 0.2931	0.8979 ± 0.0736	0.8951 ± 0.0648
Weighted sum	0.8693 ± 0.0516	0.8663 ± 0.0596	0.9247 ± 0.0132	0.9237 ± 0.0120
Concatenation	0.8841 ± 0.0290	0.8790 ± 0.0324	0.9227 ± 0.0095	0.9218 ± 0.0098
Cross-attention	0.8829 ± 0.0300	0.8246 ± 0.1755	0.9210 ± 0.0115	0.9238 ± 0.0114
Temporal only	0.8706 ± 0.0482	0.8706 ± 0.0482	0.9233 ± 0.0142	0.9233 ± 0.0142

present, and with it removed from the node features—so that the difference measures what the leak is worth.

2) *The finding replicates:* On Hamburg with the leak removed, none of the twenty paired comparisons—ten within each graph construction—is significant after Holm correction, exactly as on the original data. One comparison is now *equivalent* rather than merely undetermined: the weighted-sum hybrid against the temporal-only control differs by -0.0014 F1 with interval $[-0.0214, +0.0187]$, inside the pre-specified bound of 0.02. This is the first point in the study where the spatial branch’s contribution can be bounded rather than only left unresolved, and it comes from the network the model was not designed around.

3) *What the leaked feature actually does:* The leak does not inflate accuracy. On Kocaeli it changes nothing at all: the mean shift across the four configurations with a spatial branch is -0.0005 F1, and the temporal-only control moves by exactly zero on every run, as it must, since the graph cannot reach it.

On Hamburg the same feature is damaging. The proposed gated hybrid falls from 0.8664 ± 0.0383 to 0.6435 ± 0.2931 —a sevenfold increase in spread and one collapsed run out of ten—and cross-attention falls from 0.8829 to 0.8246. The two fixed-weight operators and the control are unaffected.

The asymmetry has a measurable explanation. Signal freeze is perfectly visible in the speed channel in both networks: consecutive speeds are identical in every instance, $P(\text{frozen} \mid \text{anomaly } 9) = 1.0000$. What differs is the converse. In Kocaeli a frozen reading implies the label with probability 0.7686, so the temporal branch already has the class and the tunnel attribute is redundant. In Hamburg the same reading implies it with probability 0.1044, so the attribute carries something the temporal branch does not: the only reliable indicator of a rare class.

That is precisely the condition under which Section VII-F1 predicts trouble. The operators that learn a mixing weight are pulled towards a spatial branch holding one perfect but rare signal, fail to escape to temporal dominance, and lose the run. The operators with a fixed weight cannot be pulled, and are not harmed.

The lesson generalises beyond this dataset. A label that leaks through a graph feature is harmless when the same label is redundantly visible to the other branch, and harmful when it is not—and the harm appears as training instability rather than as inflated accuracy. An evaluation that looked only at mean F1 on Kocaeli would have found nothing wrong, and one that looked only at mean F1 on Hamburg would have concluded

that the gating mechanism is unstable, when what is unstable is the combination of that mechanism with a leak. Neither reading is available without running both settings deliberately.

4) *Four datasets, one conclusion:* **Nothing is resolvable in any of them.** Across the four datasets, zero of the forty pairwise comparisons is significant after Holm correction. What changes between them is how tightly the null can be bounded: ten of ten comparisons are equivalent at $\delta = 0.02$ on the cleanest dataset, six on the next, two on the third, and none on the noisiest. The conclusion is the same everywhere; the precision with which it can be stated is not.

The collapse pattern holds a third time. Six runs collapse in the noisiest dataset, and all six belong to the two operators that learn a mixing weight—6 of 80 against 0 of 120 across all four datasets combined. This now holds on the original data, on a regenerated Kocaeli, and on an independent network, under two fill strategies.

A caution about reading across columns. The two fill strategies were not applied to one dataset; each was generated with its own trajectories, and because the speed-limit violation label depends on the speed limit, the labels differ between them as well. The F1 difference between a dictionary-filled and a KNN-filled dataset therefore cannot be read as an imputation effect. What is comparable is the set of within-dataset comparisons, which is what the table reports. The controlled imputation experiment is the one in Section VII-J, where four treatments are applied to the same data.

5) *What this does not establish:* Hamburg is a different road network, not a different generator: the same simulator, the same anomaly taxonomy, and the same kinematic definitions produced it. The replication therefore rules out an explanation specific to Kocaeli’s topology, and does not rule out one specific to how these anomalies are defined. Two of the delivered axes could not be used for the sensitivity they were intended to supply—the nominal prevalence settings of 1%, 5% and 15% yield realised rates averaging 16.4%, 20.8% and 24.7%, all above the rate of the original data, and the noise multiplier produces a spread across its three levels of 0.0353 against a within-level standard deviation of 0.4403—the three levels sit closer together than runs within a single level, by a factor of 12. The prevalence and noise sensitivities reported in Section VII-J are therefore derived from the original data rather than from these.

Table 32. [Cross-validation, 5 folds $\times$ 2 seeds; leaking attribute removed in every cell] Four datasets: two road networks and two strategies for filling the missing road attributes. Verdicts are over the ten pairwise comparisons within each dataset. The last two columns count degenerate runs among the operators that learn a mixing weight and among those that do not.

Dataset	Best F1	Different	Equiv.	Undet.	Collapse, learned	Collapse, fixed
Kocaeli, dictionary fill	0.9398	0	10	0	0/20	0/30
Kocaeli, KNN fill	0.9247	0	6	4	0/20	0/30
Hamburg, dictionary fill	0.8334	0	0	10	6/20	0/30
Hamburg, KNN fill	0.8841	0	2	8	0/20	0/30

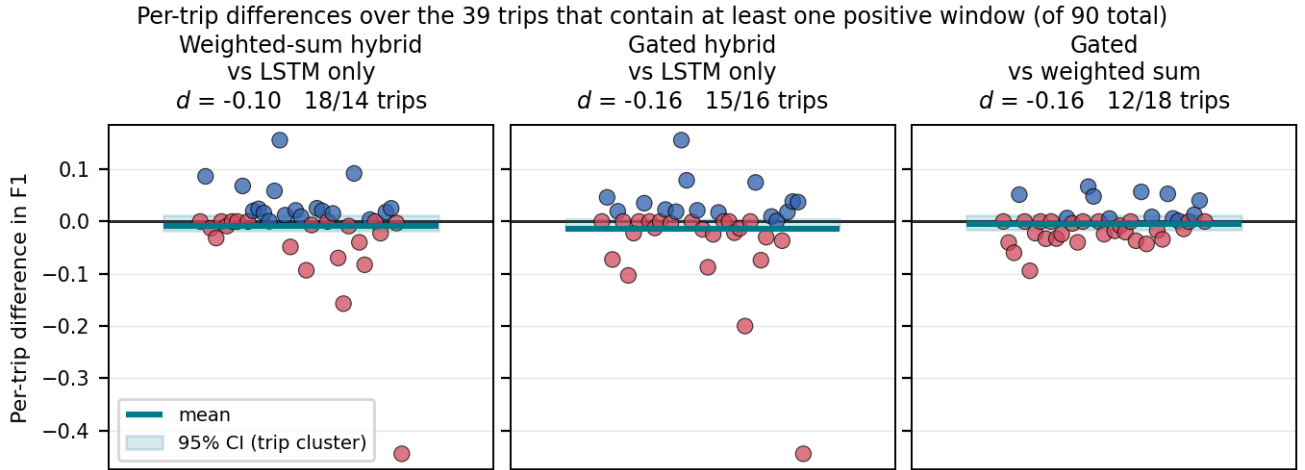


Figure 8. Per-trip differences in F1 for the three principal comparisons, over the 39 trips that contain at least one positive window. Each point is one trip; the horizontal band is the trip-level cluster-bootstrap interval for the pooled difference. The differences are centred near zero with trips falling on both sides, and each comparison has one trip whose difference is an order of magnitude larger than the rest.

M. Effect Sizes and the Distribution Across Trips

A pooled difference is a single number, and a reader is entitled to ask whether it summarises a consistent effect or averages over trips that disagree. Fig. 8 shows the per-trip differences behind the three comparisons that matter most, and Table 33 gives standardised effect sizes for them.

Two features of the design constrain what this can show. Only 39 of the 90 trips contain at least one positive window, so a per-trip F1 is defined for those 39; the remaining 51 contribute false positives but no recall, and are reported separately in the result files. The effective sample for any trip-level statement about detection quality is therefore smaller than the trip count alone suggests.

Table 33. [Cross-validation, pooled, per-trip] Standardised effect sizes over the 39 trips containing at least one positive window. Cohen’s d is computed on the paired per-trip differences. The final column counts trips favouring the first model against the second, with ties omitted.

Comparison	Mean Δ	s.d.	Cohen’s d	Trips for / against
Weighted sum vs. LSTM only	−0.0092	0.0885	−0.10	18 / 14
Gated vs. LSTM only	−0.0146	0.0899	−0.16	15 / 16
Gated vs. weighted sum	−0.0054	0.0330	−0.16	12 / 18

All three effect sizes are small by any conventional reading: $|d| \leq 0.16$. More informative than the magnitude is the split. For the weighted-sum hybrid against the temporal-only control, 18 trips favour the hybrid and 14 favour the control, with 7 tied. A pooled mean of −0.0092 is not a small consistent advantage; it is the average of a distribution straddling zero. The single

trip at −0.4444 contributes more to that mean than the median trip does, which is the behaviour the trip-level resampling of Section VI-B is designed to account for.

N. Calibration

Probability calibration was not assessed in the previous version of this work. Table 17 and the reliability diagram in Fig. 9 report it. The best-calibrated hybrid is the weighted-sum variant (ECE 0.0409), ahead of concatenation (0.0560), gating (0.0579), and cross-attention (0.0627); the temporal-only control attains the lowest Brier score (0.0208).

We note this as a correction to the earlier version of this work, which reported the gated operator as clearly best calibrated (ECE 0.0338 against 0.0594 for the weighted sum) on the basis of a single seed per fold. With two seeds per fold the ordering of the two is reversed and the gap is small. Calibration differences of this magnitude are not separable from run-to-run variation at this sample size, and we no longer claim calibration as an advantage of the gating mechanism.

The general point survives the correction even though the specific claim does not. In a monitoring context where alerts are ranked or thresholded by predicted risk, a better-calibrated score is more useful than a marginally higher F1, and calibration is worth reporting for that reason regardless of which variant wins. All five configurations are mildly overconfident, with ECE between 0.041 and 0.063; none is well enough calibrated to be used as a risk score without post-hoc recalibration.

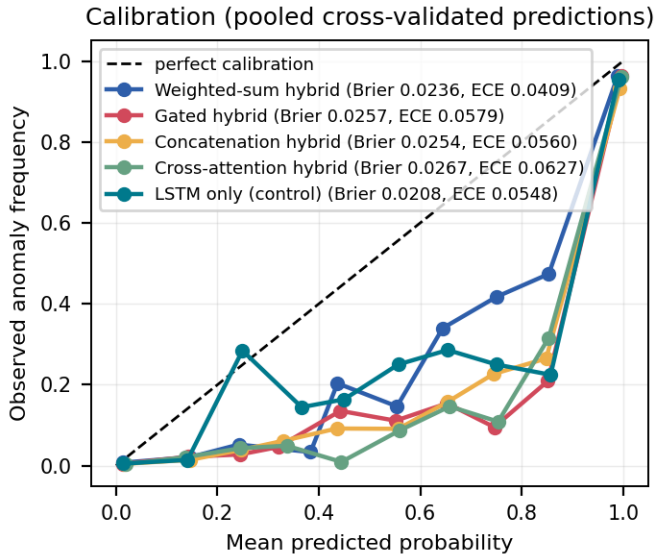


Figure 9. Reliability diagram from pooled cross-validated predictions.

O. Graph Construction Ablation

Section V-B noted that the graph can be built from trip-specific segment instances (4,678 nodes) or merged by physical road (628 nodes). Table 34 reports both.

Table 34. [Fixed partition, 5 seeds] Effect of graph construction, five seeds. The fusion operator is named explicitly in every row; an earlier version of this table labelled its hybrid rows only as “hybrid”, which made them appear inconsistent with the fusion ablation of Table 15 when in fact a different operator had been used.

Configuration	Fusion	Nodes	Deg.	F1-Score	ROC-AUC
GAT only, trip-instance	—	4,678	11.47	0.0767 ± 0.0035	0.5910
GAT only, merged	—	628	1.11	0.0745 ± 0.0047	0.7115
Hybrid, trip-instance	weighted sum	4,678	11.47	0.8713 ± 0.0188	0.9638
Hybrid, merged	weighted sum	628	1.11	0.8413 ± 0.0010	0.9228
Hybrid, trip-instance	gated	4,678	11.47	0.6953 ± 0.3488	0.8752
Hybrid, merged	gated	628	1.11	0.8493 ± 0.0274	0.9684

Merging duplicate nodes substantially improves the *standalone* discriminative power of the spatial branch (ROC-AUC 0.5910 → 0.7115), which is consistent with the duplicate structure degrading message passing. Its F1 nonetheless remains near 0.07, consistent with the label-ambiguity ceiling of Section V-B: several sequences with different labels map to the same node, so no threshold yields good precision and recall simultaneously.

Neither construction is designated primary (Section V-B); both are reported throughout. With the fusion operator held fixed, the hybrid is stronger on the trip-instance graph (0.8713 versus 0.8413 for weighted sum), consistent with its higher per-node label purity. The row that repays closer reading is the last one. **The gated operator does not collapse on the merged graph:** it attains 0.8493 ± 0.0274 there, against 0.6953 ± 0.3488 on the trip-instance graph, and records the highest ROC-AUC in the table. This locates the collapse mechanism precisely. The gate fails when it locks onto a spatial branch that carries almost no standalone signal (ROC-AUC 0.5910, barely above chance); on the merged graph the same branch reaches 0.7115,

and locking onto it is no longer catastrophic. The instability of Section VII-C is therefore not a property of gating in the abstract but of gating over an uninformative modality—which is a sharper and more transferable statement than the one we could make from the fixed graph alone.

P. Statistical Reassessment at the Trip Level

Every p -value reported so far treats each of the 4,799 test windows as an independent observation. Section VI-B explained why that assumption does not hold: windows overlap by 80%, and all windows of a trip share a driver, a vehicle, and a route. This subsection repeats the same comparisons with the trip as the unit and reports what changes. It is placed here, after the results it revises, because the revision applies to all of them at once.

The fixed test partition contains 4,799 windows drawn from **14 trips** (between 75 and 760 windows each, mean 343). Fourteen is the effective sample size for any inference on this partition.

Table 35. [Fixed partition, window- and trip-level] The same comparisons at the window level and at the trip level. $\Delta F1$ and intervals are computed on the median seed. Trip-level p -values come from sign-flip permutation over trips; the Holm column corrects within each family. Widening is the ratio of trip-level to window-level interval width.

Family	Comparison	p (window)	p (trip)	Holm	Widen.
Baselines	vs. LSTM only	0.549	0.749	1.000	2.43×
	vs. LightGBM	0.302	0.219	0.657	1.81×
	vs. XGBoost	0.049	0.064	0.320	1.80×
	vs. Random Forest	0.754	0.632	1.000	1.07×
	vs. threshold rule	0.000	0.065	0.320	1.84×
Fusion	gated vs. weighted sum	0.031	0.032	0.097	1.25×
	gated vs. concatenation	1.000	0.500	0.755	1.16×
	gated vs. cross-attention	0.727	0.377	0.755	0.85×
Encoders	vs. adaptive adjacency	0.000	0.007	0.056	2.71×
	vs. gated TCN	0.003	0.048	0.335	1.44×
	vs. Chebyshev conv	0.007	0.061	0.365	2.79×
	vs. diffusion conv	0.003	0.062	0.365	2.22×
	vs. GRU	0.118	0.375	1.000	1.71×
	vs. dilated conv	0.115	0.503	1.000	1.19×
	vs. Transformer	0.815	0.439	1.000	1.17×
	vs. no spatial branch	0.549	0.748	1.000	2.43×

The outcome is unambiguous and we report it without softening. **Under the correct statistical unit, no comparison on the fixed partition remains significant after multiplicity correction.** Seven comparisons that were significant at the window level are not significant at the trip level: the advantage over XGBoost, the advantage over the zero-parameter threshold rule, the gated-versus-weighted-sum difference, and the margins over the STGCN gated temporal convolution, Chebyshev convolution, diffusion convolution, and adaptive adjacency. Confidence intervals widen by factors of 1.07 to 2.79. The hybrid model’s own F1 on this partition, 0.8851 on the median seed, carries a trip-level interval of [0.7231, 0.9470].

Two of these deserve comment rather than a bare verdict. The comparison against adaptive adjacency has a raw trip-level p of 0.007 and a cluster-bootstrap interval of [+0.679, +0.944] that excludes zero by a wide margin; it fails only after Holm correction across eight comparisons, at $p = 0.056$. Given an

effect size of +0.86 F1, we regard the collapse of adaptive adjacency as established and its formal non-significance as a statement about the correction and the cluster count rather than about the effect. Conversely, the gated-versus-weighted-sum difference has a raw p of 0.032 but an effect size of -0.038 , and it does not survive correction; we do not claim it from this partition.

The general lesson is the one this table was constructed to expose. A literature that evaluates overlapping windows as independent observations on a single partition will report significance that a correctly specified test does not support. That is not a criticism of any particular prior study; it applies with full force to the earlier version of this work, whose central ablation claim rested on a window-level p of 0.035 from one partition. What remains, and what the following subsection assesses, is the cross-validated evidence—in which all 90 trips receive an out-of-fold prediction rather than 14.

Q. Computational Efficiency and Edge Feasibility

Table 36 and Fig. 10 report parameter counts and single-window inference latency measured with batch size 1, after 50 warm-up iterations, over 500 repetitions on GPU and 60 on CPU. Measurements are taken on the **trip-instance graph** (4,678 nodes), the larger of the two constructions and therefore the upper bound on the spatial branch’s inference cost; the merged graph is cheaper by construction. The previous version reported latency measured on the merged 628-node graph while reporting accuracy on the trip-instance graph, which understated the cost of the configuration actually being evaluated; the figures below supersede those.

Table 36. [Single-window latency, no training] Single-window inference latency (median) and parameter counts, measured on the trip-instance graph (4,678 nodes)

Model	Params	GPU	GPU cached	CPU	CPU cached
Hybrid (weighted sum, GAT)	220,802	2.87 ms	0.64 ms	27.82 ms	7.03 ms
Hybrid (gated, GAT)	224,962	2.80 ms	0.78 ms	25.07 ms	7.63 ms
Hybrid (concat)	222,849	2.56 ms	0.55 ms	23.14 ms	7.32 ms
Hybrid (cross-attention)	237,441	2.91 ms	0.96 ms	19.83 ms	6.61 ms
LSTM only	215,234	0.80 ms	0.64 ms	7.69 ms	7.03 ms
GRU + GAT	170,626	2.60 ms	0.66 ms	33.63 ms	8.58 ms
Gated TCN + GAT	122,242	2.76 ms	0.88 ms	24.79 ms	1.83 ms
Dilated conv + GAT	416,258	4.22 ms	2.26 ms	27.42 ms	5.98 ms
Transformer + GAT	292,098	3.30 ms	1.30 ms	22.48 ms	2.84 ms
Adaptive adjacency + LSTM	321,850	6.18 ms	0.69 ms	55.40 ms	6.85 ms

The primary model contains 220,802 parameters (≈ 0.9 MB in fp32), of which the temporal branch accounts for 200,704 (90.9%), the projection layers 12,416, the classifier 2,113, the spatial branch only 5,568 (2.5%), and the fusion operator a single scalar. Peak GPU memory is 66 MB.

Despite that small share, the spatial branch dominates inference cost, because a naive implementation recomputes embeddings for all 4,678 nodes on every forward pass. **Since the road network is static, node embeddings can be computed once and cached**, reducing GPU latency from 2.87 to 0.64 ms ($4.5\times$) and CPU latency from 27.82 to 7.03 ms ($4.0\times$). After caching, the hybrid model matches the temporal-only model exactly on both devices (0.64 ms on GPU, 7.03 ms on CPU): once the graph forward pass is amortized, spatial

context is free at inference time. This holds for every fusion operator, since all four cost at most 16,640 parameters and none appears in the latency ranking.

At 27.82 ms per uncached window on CPU ($p_{95} = 39.92$ ms), or 7.03 ms cached ($p_{95} = 8.77$ ms), a single-core edge device could process on the order of 35 windows per second without caching and 140 with it, against a one-second sampling resolution. Caching is therefore not an optimization but a precondition for single-core deployment at scale. Memory is not a constraint at this model size. These figures characterize computational feasibility only; they do not address detection quality under real sensing conditions.

R. Scope of the Contribution

It is worth stating directly what kind of claim this paper does and does not make, because the two are easily conflated in this literature.

This is an *evaluation study*. Its object is a set of design decisions—encoder pairing, fusion operator, graph construction—and its result is a statement about how much evidence a controlled comparison can actually support for each of them. It is not a report of a deployable traffic anomaly detector, and none of the numbers here should be read as an estimate of field performance. The data are simulated; the labels are, for most subtypes, threshold functions of the model’s own inputs; the road network is one city; the trip count is 90. Each of these independently rules out a deployment claim, and we make none.

What survives those limitations are the *relative* statements, and they survive precisely because every variant faces the identical task, partition, budget, and decision rule. Three of them are worth separating. A sample-wise gate presented as the architectural contribution of a hybrid model does not, on this data, outperform a single learnable scalar, and the gap between them is smaller than the variation across random seeds—indeed smaller than the variation across repeated runs of either one. The ordering of the four fusion operators is not a property of the operators at all under this protocol: it changes three times as the number of averaged seeds goes from five on one partition, to two under cross-validation, to five. And the one performance difference the study can resolve at the trip level runs *against* the proposed architecture, with gradient boosting on hand-crafted window aggregates ahead of the gated hybrid by 0.0409 F1 (Holm $p = 0.0002$). We state this last result in the abstract rather than in a footnote, because a study whose argument is that evaluation protocol determines conclusions cannot be selective about which of its own conclusions it reports. None of the three depends on the data being real; all three depend on the comparison being controlled, which is why Section VI-A4 and Section VI-B are given the space they are.

Whether the same conclusion holds on field-collected trajectories is an open question that this dataset cannot answer, and we identify it in Section VIII-C as the decisive next experiment rather than implying it here.

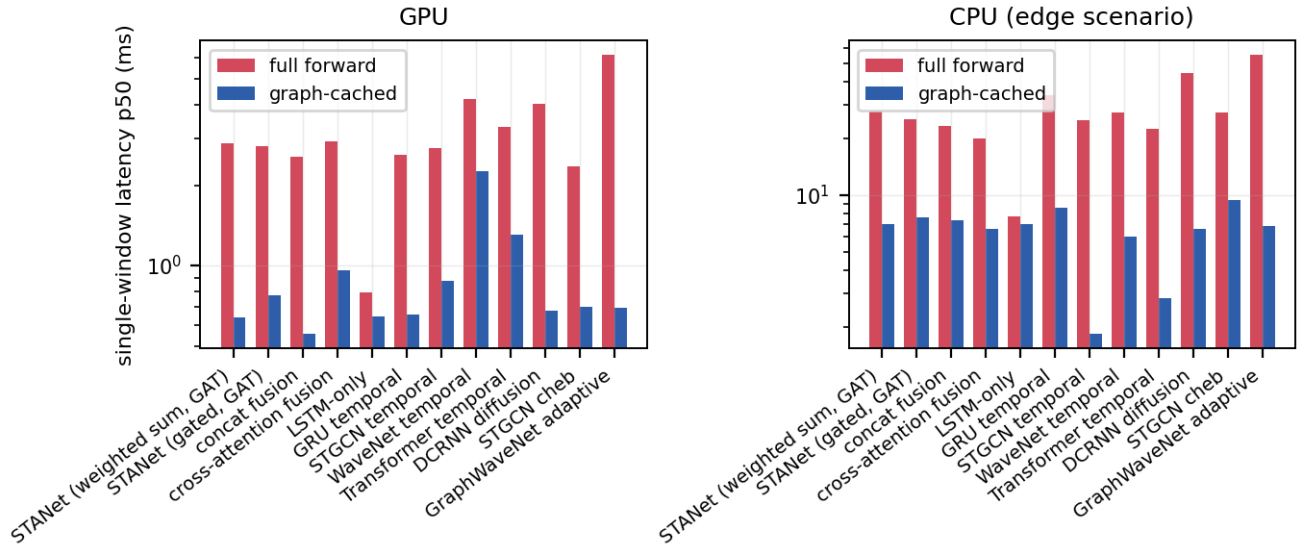


Figure 10. Single-window latency with and without graph-embedding caching, on GPU and CPU (log scale).

S. Threats to Validity

Synthetic data. All findings are obtained on simulated trajectories. No real sensor data, verified incident record, or prospective stream is involved, and no claim about real-world detection performance is made.

One generator. The four datasets of Section VII-L4 span two road networks and two imputation strategies, but a single simulator produced all of them, with the same anomaly taxonomy and the same kinematic definitions. The replication therefore eliminates an explanation specific to one topology; it does not eliminate one specific to how these anomalies are defined. Since most subtypes are threshold functions of the kinematic channels, a generator whose anomalies depended more strongly on road context might well give the spatial branch something to do. That experiment is not in this paper, and it is the one we would run next.

Generator parameters. The transition probabilities behind the behavioural model were not available to us. Section V-B1 recovers the realised transition frequencies from the released trajectories so that the data’s behaviour is documented, but how those probabilities were chosen or calibrated is not, and a reader cannot regenerate an equivalent dataset from the description alone.

Label-input dependence. Most anomaly subtypes are threshold functions of the model’s own inputs (Section VII-A). Conclusions about *relative* architectural merit are unaffected, since all configurations face the same task, but absolute performance figures reflect a partly self-referential problem and should not be compared to results on independently labelled data.

Statistical power. This is the tightest constraint on what the paper can claim, and it is a consequence of the sampling design rather than of the analysis. Because windows overlap by 80% and share a trip, the unit of analysis is the trip (Section VI-B), and the dataset contains only 90 trips: roughly 13 of them in the fixed test partition, all 90 across the cross-validation

folds. Interval estimates on the fixed partition are therefore wide, and comparisons that appear significant at the window level frequently are not once the correct unit is used. We report both, labelled, and base conclusions on the trip-level, Holm-corrected values from the cross-validated analysis. Differences that the design cannot resolve are presented as point estimates with intervals and are not claimed as significant—including differences that would favour the architecture proposed in the earlier version of this work.

Encoder adaptation. STGCN, DCRNN, and Graph WaveNet are represented by their core operators rather than their full published architectures. This is necessary given the task mismatch, but the results should be read as evidence about those operators in this setting rather than as a benchmark of the published systems.

Imputation dependence. Roughly half of the lane counts and speed limits are author-defined defaults, which directly affects the node features available to the spatial branch. Four treatments of those defaults were compared (Section VII-J) and none differs measurably, including removing the two attributes entirely; a second fill strategy on regenerated data behaves the same way (Section VII-L4). The dependence is therefore bounded rather than eliminated: attributes that carry no measurable signal cannot be sensitive to how they were filled, and the bound would have to be re-established on data where they do carry signal.

A label determined by a graph feature. In the datasets delivered for this revision, every record on a tunnel segment carries the signal-freeze label, and the prescribed preprocessing enforces this by deleting the exceptions (Section VII-L1). Because the tunnel flag is a node attribute, a graph-enabled model can read that label off a single input. We removed the attribute and report both settings, but the condition is a property of the released data rather than something we introduced, and any reuse of those datasets should handle it.

Deleted counterexamples. The same prescribed filter removes records that exceed the speed limit without carrying the

corresponding label, rather than relabelling them. This makes the published generation rule exactly true of the cleaned data by construction, which overstates how deterministic the labelling function is (Section V-D1).

VIII. CONCLUSION AND FUTURE WORK

This study evaluated the components of a hybrid LSTM–GAT traffic anomaly detector under a controlled protocol in which one element is varied at a time and every result is reported over multiple seeds with interval estimates and paired tests.

A. Summary of Findings

- **The fixed partition supports no claim at all.** It holds 4,799 windows but 14 trips. At the trip level, after Holm correction, none of sixteen comparisons is significant—including seven that are significant when windows are treated as independent. The hybrid model’s own F1 on that partition carries an interval of $[0.7231, 0.9470]$.
- **The fusion ranking is partition-dependent, in both directions.** The sample-wise gate ranks last of four on the fixed partition and first of four under cross-validation, where no operator differs significantly from any other. We therefore make no accuracy claim for or against the gating mechanism, and we withdraw the earlier claim against it.
- **A reproducible collapse mode belongs to a class of operators, not to one of them.** The gate collapsed to $F1 = 0.0715$ at one of five seeds on the fixed partition, recurring at the same seed in an independent re-run under a different backend configuration. Over 500 repeated-partition runs the rate is 4.0% for the two operators that learn a mixing weight and 0.0% for the three that do not, and it is not confined to one graph construction: the gate collapses in 3 of 50 runs on the merged graph against 1 of 50 on the trip-instance graph, the opposite of what the five-seed fixed partition suggested. Epoch-level measurement identifies the mechanism. This is a failure mode practitioners should know about when a fusion stage learns its own weighting—not evidence of general inferiority of the proposed operator.
- **No benefit from the spatial branch is demonstrated.** Under cross-validation no anomaly subtype shows a significant gain from the spatial branch after correction, and pooled F1 is highest for the control that has no spatial branch at all. This is an absence of demonstrated benefit, not a demonstration of absence: where the distinction matters we test it directly (Section VI-D) rather than reading it off a large p -value. A gain on abnormal-dwell detection reported in the previous version was an artifact of a pooled threshold and is withdrawn (Section VII-E1); the comparison now closest to significance has the hybrid performing *worse* on signal-freeze events.
- **Encoder choice separates along one axis only.** Graph attention ranks above Chebyshev convolution, diffusion convolution, and adaptive adjacency, and the separation is clearest against adaptive adjacency, which is the only one of the three that the fixed-partition design resolves

after correction (Section VI-D); the LSTM, a GRU, and a Transformer encoder finish within 0.0015 F1 of one another. The spatial axis is the one along which differences appear at all; the recurrence axis is not.

- **Gate interpretability is seed-dependent.** The class-conditional shift in g_{gate} varies in magnitude and sign across seeds and does not constitute a stable model property.
- **Deployment cost is removable** (Section VII-Q). Caching static node embeddings reduces single-window latency by $4.5\times$ on GPU and $4.0\times$ on CPU, after which the hybrid model matches temporal-only latency exactly.
- **Simple references are strong.** A zero-parameter three-threshold rule reaches $F1 = 0.8536$ and gradient boosting on window aggregates matches the neural model to within 0.0014 F1, with higher ROC-AUC.

B. Implications

The limitations bounding these conclusions are collected in Section VII-S. The broader implication is methodological, and it has three parts that we would separate.

The unit of analysis is not a detail. Sliding-window evaluation is near-universal in this literature, and windows cut with a stride shorter than their length are not independent observations. Reporting window-level p -values under that design does not slightly overstate significance; in our case it turned sixteen non-significant comparisons into seven significant ones. Any study using overlapping windows should state its cluster structure and test at the level of the independent unit—here the trip, elsewhere the vehicle, the sensor, or the day.

One partition cannot order architectures. We report the same four fusion operators ranked in opposite orders on one partition and across five, with the mechanism under study finishing both last and first. Neither ordering is wrong; both are under-determined. A single grouped partition, even a well-constructed one, is a sample of size one from the space of partitions, and architectural rankings are not stable with respect to that sample.

Error bars must be earned. Extending our encoder benchmark from three seeds to five raised the leading configuration’s standard deviation from 0.0033 to 0.0188 and removed two of its four significant margins. A tight standard deviation over three runs is not evidence of stability; it is a small sample of a quantity whose spread has not been measured.

We would additionally encourage reporting a rule-based or classical reference whenever labels are generated by known rules, since it bounds how much of the measured performance is attributable to learning at all—in this study, a zero-parameter rule reaches within 0.018 F1 of every neural configuration tested.

C. Future Directions

- **Dataset redesign.** Sharing road segments across trips and increasing the prevalence of context-dependent subtypes such as speed-limit violation would allow the spatial branch’s contribution to be tested with adequate power—the decisive experiment that the present data cannot support.

- **Simulation-to-real transfer.** Findings obtained on physics-informed synthetic trajectories require domain adaptation before transfer to field data. Digital twin-assisted graph contrastive domain adaptation has shown promise for adapting simulation-derived representations to small-sample target domains [6]; adding such a layer to the spatial branch is a natural next step for cross-city transfer.
- **Stabilized gating.** Since the collapse mode arises from the multiplicative form of Eq. 16, entropy regularization on g_{gate} , a bounded gating range, or an auxiliary supervision signal on the gate are candidate remedies worth evaluating.
- **Imputation sensitivity.** Road-type-specific lane and speed-limit defaults should be varied systematically to quantify how strongly the spatial branch depends on author-defined assumptions.
- **Real-world and cross-city validation.** Field-collected sensor data, independently verified incidents, prospective streams, and external road networks are required before any claim of deployment suitability.

DATA AND CODE AVAILABILITY

The synthetic dataset, its data card, the complete preprocessing and modelling code, the experiment scripts, every result file reported in this paper, and the script that verifies the manuscript's numbers against those files are available at <https://github.com/keremkartal/Anomaly-Detection> and permanently archived at doi:10.5281/zenodo.22637361, which resolves to the exact snapshot (v2.0.0) that produced every number reported here. The concept identifier doi:10.5281/zenodo.22637360 resolves to the most recent version. No real-world vehicle trajectory data was used.

The repository contains two mechanisms that support the internal consistency of the tables reported here. First, every result file carries a *protocol stamp*: a hash over the seed set, epoch budget, patience, batch size, learning rate, scheduler, dropout, checkpoint criterion, threshold rule, cuDNN flags, window length, stride, split seed, and library versions. Scripts that consume another script's results assert that the stamps match and abort otherwise. Second, configurations that appear in more than one table are trained once and cached under their protocol stamp, so those tables read the same run rather than two independently trained runs that are assumed to agree. Section VI-A4 describes why these mechanisms were introduced.

Because several anomaly subtypes are defined as threshold functions of the model's input channels, a zero-parameter rule-based classifier is reported alongside all learned models as a reference point.

REFERENCES

- [1] X. Huang, N. Chen, Z. Deng, and S. Huang, "Multivariate time series anomaly detection via dynamic graph attention network and Informer," *Applied Intelligence*, vol. 54, no. 17-18, pp. 7636–7658, Sep. 2024. [Online]. Available: <https://link.springer.com/10.1007/s10489-024-05575-y>
- [2] M. Duan, S. Sun, and M. Liu, "A multimodal deep fusion framework for highway traffic anomaly detection," *Scientific Reports*, vol. 15, p. 33573, Sep. 2025. [Online]. Available: <https://www.nature.com/articles/s41598-025-18671-x>
- [3] H. D. Nguyen, A. T.-H. Nguyen, and T.-H. Do, "Spatiotemporal traffic prediction in Dublin: Volume, speed, and influencing factors," *Engineering Applications of Artificial Intelligence*, vol. 166, p. 113602, Feb. 2026. [Online]. Available: <https://linkinghub.elsevier.com/retrieve/pii/S0952197625036346>
- [4] Q. Liang, L. Cong, and H. Du, "Anomaly traffic detection in heterogeneous lightweight networks based on spatio-temporal features," *The Journal of Supercomputing*, vol. 81, no. 5, p. 690, Apr. 2025. [Online]. Available: <https://link.springer.com/10.1007/s11227-025-07189-8>
- [5] H. Cui, K. Xiao, H. Wang, and X. Zhang, "Lane-flow-learning based autonomous vehicle trajectory prediction using spatial-temporal fusion attention," *Information Sciences*, vol. 737, p. 123134, May 2026. [Online]. Available: <https://linkinghub.elsevier.com/retrieve/pii/S0020025526000654>
- [6] C. Dai, D. He, Z. Jin, X. Zhang, G. Chen, J. Wu, J. Zhao, and Y. Xu, "Digital twin-assisted graph contrastive domain adaptation for small-sample bearing fault diagnosis," *Structural Health Monitoring*, Apr. 2026, art. no. 14759217261440798. [Online]. Available: <https://doi.org/10.1177/14759217261440798>
- [7] H. Zhang, S. Zhao, R. Liu, W. Wang, Y. Hong, and R. Hu, "Automatic Traffic Anomaly Detection on the Road Network with Spatial-Temporal Graph Neural Network Representation Learning," *Wireless Communications and Mobile Computing*, vol. 2022, no. 1, p. 4222827, Jan. 2022. [Online]. Available: <https://onlinelibrary.wiley.com/doi/10.1155/2022/4222827>
- [8] Y. Zhang, S. Yang, H. Wang, Y. Cheng, J. Wang, L. Cao, and Z. An, "A traffic flow forecasting method based on hybrid spatial-temporal gated convolution," *International Journal of Machine Learning and Cybernetics*, vol. 16, no. 3, pp. 1805–1817, Mar. 2025. [Online]. Available: <https://link.springer.com/10.1007/s13042-024-02364-4>
- [9] Z. Ren, X. Li, J. Peng, K. Chen, Q. Tan, X. Wu, and C. Shi, "Graph autoencoder with mirror temporal convolutional networks for traffic anomaly detection," *Scientific Reports*, vol. 14, no. 1, p. 1247, Jan. 2024. [Online]. Available: <https://www.nature.com/articles/s41598-024-51374-3>
- [10] H. Li, Y. Ma, X. Wang, and Z. Li, "Graph Spatiotemporal Pattern Learning Network for Real-Time Road Network Traffic Abnormal Incident Detection," *Transportation Research Record: Journal of the Transportation Research Board*, vol. 2677, no. 12, pp. 815–829, Dec. 2023. [Online]. Available: <https://journals.sagepub.com/doi/10.1177/03611981231170004>
- [11] D. Park, S.-S. Choi, D. Lim, and Y.-S. Kang, "Graph Multi-Resolution Transformer for Road Traffic Anomaly Detection," *IEEE Access*, vol. 13, pp. 27 428–27 437, 2025. [Online]. Available: <https://ieeexplore.ieee.org/document/10872917>
- [12] J. Chen, H. Chen, and H. Lu, "Enhancing Road Safety and Sustainability: A Multi-Scale Temporal Model for Vehicle Trajectory Anomaly Detection in Road Network Interactions," *Sustainability*, vol. 18, no. 2, p. 597, Jan. 2026. [Online]. Available: <https://www.mdpi.com/2071-1050/18/2/597>
- [13] R. Zhong, B. Hu, F. Wang, Y. Feng, Z. Li, X. Song, Y. Wang, S. Lou, and J. Tan, "Multi-factor embedding GNN-based traffic flow prediction considering intersection similarity," *Neurocomputing*, vol. 620, p. 129193, Mar. 2025. [Online]. Available: <https://linkinghub.elsevier.com/retrieve/pii/S0925231224019647>
- [14] C. Pan, J. Zhu, Z. Kong, H. Shi, and W. Yang, "DC-STGCN: Dual-Channel Based Graph Convolutional Networks for Network Traffic Forecasting," *Electronics*, vol. 10, no. 9, p. 1014, Apr. 2021. [Online]. Available: <https://www.mdpi.com/2079-9292/10/9/1014>
- [15] J. Chang, J. Yin, Y. Hao, and C. Gao, "STFDSGCN: Spatio-Temporal Fusion Graph Neural Network Based on Dynamic Sparse Graph Convolution GRU for Traffic Flow Forecast," *Sensors*, vol. 25, no. 11, p. 3446, May 2025. [Online]. Available: <https://www.mdpi.com/1424-8220/25/11/3446>
- [16] L. Zhou, Q. Zeng, and B. Li, "Hybrid Anomaly Detection via Multihead Dynamic Graph Attention Networks for Multivariate Time Series," *IEEE Access*, vol. 10, pp. 40 967–40 978, 2022. [Online]. Available: <https://ieeexplore.ieee.org/document/9758699>
- [17] I. Taheri Sarteshnizi, M. Sarvi, S. Asadi Bagloee, and N. Nassiri, "On the applicability of time series anomaly detection methods to real-world traffic volume data," *Transportation Research Part C: Emerging Technologies*, vol. 184, p. 105536, Mar. 2026. [Online]. Available: <https://linkinghub.elsevier.com/retrieve/pii/S0968090X26000240>
- [18] C. Hu, Y. Wang, X. Zhang, M. Zheng, G. Feng, J. Liu, and Y. Kang, "Dynamic Graph Neural Network-Transformer-LSTM Based Multi-Scale Spatio-Temporal Traffic Forecasting," *International Journal of Intelligent Transportation Systems Research*, Feb. 2026. [Online]. Available: <https://link.springer.com/10.1007/s13177-025-00623-4>
- [19] L. Zhu, Q. Zhang, X. Jian, Y. Yang, and L. Li, "Spatio-temporal traffic

- accidents detection via graph based generative adversarial network,” *Engineering Applications of Artificial Intelligence*, vol. 165, p. 113488, Feb. 2026. [Online]. Available: <https://linkinghub.elsevier.com/retrieve/pii/S0952197625035195>
- [20] D. Xia, B. Shen, J. Geng, Y. Hu, Y. Li, and H. Li, “Attention-based spatial-temporal adaptive dual-graph convolutional network for traffic flow forecasting,” *Neural Computing and Applications*, vol. 35, no. 23, pp. 17 217–17 231, Aug. 2023. [Online]. Available: <https://link.springer.com/10.1007/s00521-023-08582-1>
- [21] J. Wang, K. Liu, H. Li, Q. Gao, X. Wang, and Y. Gong, “Multi- Scale Temporal Features-Based Hybrid LSTM-GAT for Traffic Flow Prediction,” *IEEE Internet of Things Journal*, pp. 1–1, 2025. [Online]. Available: <https://ieeexplore.ieee.org/document/11127181/>
- [22] J. Feng, Y. Zhang, X. Piao, Y. Hu, and B. Yin, “Traffic Anomaly Detection based on Spatio-Temporal Hypergraph Convolution Neural Networks,” *Physica A: Statistical Mechanics and its Applications*, vol. 646, p. 129891, Jul. 2024. [Online]. Available: <https://linkinghub.elsevier.com/retrieve/pii/S037843712400400X>
- [23] Y. Ma, Y. Nie, J. Guo, J. Zhu, F. Li, Q. Xie, and A. V. Vasilakos, “UTFP-AD: Urban Traffic Flow Pattern Anomaly Detection Using Spatiotemporal Data,” *IEEE Transactions on Intelligent Transportation Systems*, vol. 26, no. 10, pp. 15 970–15 984, Oct. 2025. [Online]. Available: <https://ieeexplore.ieee.org/document/11029517/>
- [24] W. Guo, W. Xu, C. Yang, Z. Zhao, X. Gao, W. Yao, and S. Jin, “CRFPI-Net: context-aware risk feature perception and inference network for pixel-level urban traffic risk mapping,” *Advanced Engineering Informatics*, vol. 71, p. 104299, Apr. 2026. [Online]. Available: <https://linkinghub.elsevier.com/retrieve/pii/S1474034625011929>
- [25] Q. He, F. Zhang, G. Bian, W. Zhang, and Z. Li, “C-lstm traffic anomaly detection model based on attention mechanism,” *Concurrency and Computation: Practice and Experience*, vol. 37, no. 25-26, p. e70314, Nov. 2025. [Online]. Available: <https://onlinelibrary.wiley.com/doi/10.1002/cpe.70314>
- [26] K. Hirata, Y. Kawasaki, and T. Yoshida, “A Traffic Anomaly Detection Method Using Traffic Flow Vectors During Heavy Rainfall,” *International Journal of Intelligent Transportation Systems Research*, vol. 23, no. 1, pp. 91–103, Apr. 2025. [Online]. Available: <https://link.springer.com/10.1007/s13177-024-00438-9>
- [27] X. Chen, G. Gu, and X. Ning, “GMS: Graph-Guided Multi-Modal Spatio-Temporal Fusion model for trajectory prediction in autonomous driving,” *Neurocomputing*, vol. 661, p. 132010, Jan. 2026. [Online]. Available: <https://linkinghub.elsevier.com/retrieve/pii/S0925231225026827>
- [28] D. Wen et al., “HDSVT: High-Density Semantic Vehicle Trajectory Dataset Based on a Cosmopolitan City Bridge,” *Scientific Data*, vol. 13, no. 1, p. 5, Dec. 2025.
- [29] J. Feng, Y. Zhang, X. Piao, Y. Hu, and B. Yin, “Traffic Anomaly Detection based on Spatio-Temporal Hypergraph Convolution Neural Networks,” *Physica A*, vol. 646, p. 129891, Jul. 2024.
- [30] B. Yu, H. Yin, and Z. Zhu, “Spatio-temporal graph convolutional networks: A deep learning framework for traffic forecasting,” in *Proc. IJCAI*, 2018, pp. 3634–3640.
- [31] Y. Li, R. Yu, C. Shahabi, and Y. Liu, “Diffusion convolutional recurrent neural network: Data-driven traffic forecasting,” in *Proc. ICLR*, 2018.
- [32] Z. Wu, S. Pan, G. Long, J. Jiang, X. Chang, and C. Zhang, “Graph wavenet for deep spatial-temporal graph modeling,” in *Proc. IJCAI*, 2019, pp. 1907–1913.
- [33] M. L. Francies, A. T. Khalil, H. M. Amer, and M. M. Ata, “A framework for continual learning in real-time traffic forecasting utilizing spatial-temporal graph convolutional recurrent networks,” *Complex & Intelligent Systems*, vol. 11, no. 10, p. 420, Oct. 2025.
- [34] L. A. Pipes, “Car following models and the fundamental diagram of road traffic,” *Transportation Research*, vol. 1, no. 1, pp. 21–29, 1967.
- [35] AASHTO, *A Policy on Geometric Design of Highways and Streets*, 2018.